2ND YEAR SYLLABUS

For academic year 2026-2027

DEPARTMENT OF MECHANICAL ENGINEERING,

SSIT

SRI SIDDHARTHA ACADEMY OF HIGHER EDUCATION

Vision:

To carve technically proficient and competent mechanical engineering graduates capable of addressing the needs of the society and the nation.

Mission:

- To impart quality technical education in core areas of Mechanical Engineering.
- To inculcate Industrial practices for better professional growth.
- To imbibe ethical values, promote entrepreneurship, impart soft skills and teamwork capabilities.
- To ensure self-learning capabilities with an aptitude for research focusing on societal needs

Program Educational Objectives [PEOs]:

PEO– 01 Provide students with sound foundation in mathematical, scientific and engineering fundamentals necessary to formulate, solve and analyze problems in the core/allied areas of Mechanical Engineering.

PEO– 02 Provide graduates with competency in the synthesis, modeling and analysis of mechanical/thermal/fluid data, assemblies and systems, research and consultancy in core areas of mechanical engineering, ensuring noticeable social impact.

PEO– 03 Promote awareness about professional ethics, inculcate skills for usage of software tools and prepare graduates of the program for continuous learning capabilities in interdisciplinary/multidisciplinary domains

Program Outcomes [POs]:

Our Engineering Graduates will be able to:

1. **Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals , engineering specialization to the solution of complex engineering problems.

2. **Problem Analysis:** Identify, formulate, review research literature, analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

3. **Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health, safety, cultural, societal and environmental considerations.
4. **Conduct Investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. **Modern Tool Usage:** Create, select, apply appropriate techniques, resources, modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The Engineer and The World:** .Apply contextual reasoning to assess societal, health, safety, legal, cultural, and environmental issues, demonstrating responsibility for sustainable development.
7. **Ethics:** Apply ethical principles, commit to professional ethics, responsibilities and norms of the engineering practice.
8. **Individual and CollaborativeTeam work:** Function effectively as an individual, as a member or leader in diverse teams in multidisciplinary settings.
9. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend , write effective reports and design documentation, make effective presentations, give and receive clear instructions.
10. **Project Management and Finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
11. **Life- Long Learning:** Recognize the need for, have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Programme Specific Outcomes (PSOs):

Our Engineering graduates are able to

PSO1: Automated/Additive Manufacturing--demonstrates knowledge ,understanding of manual and automated assembly/manufacturing systems including additive manufacturing systems.

PSO2: Entrepreneurship--recognize the need for and ability to engage in entrepreneurial activities.

SRI SIDDHARTHA ACADEMY OF HIGHER EDUCATION (SSAHE)

(Deemed to be University, Accredited A+ Grade by NAAC)

SRI SIDDHARTHA INSTITUTE OF TECHNOLOGY, TUMAKURU - 572105

(A Constituent College of SSAHE, Accredited by NBA, New Delhi in Tier - 1)

Scheme of Teaching and Examination for NEP 160 Credits (2025 Scheme)

3RD SEMESTER **ACADEMIC YEAR: 2026-27**

Sl. No.	Category and Course Code		Course Title	Teach. Dept.	Teaching Hrs. / Semester					Credits	Evaluation Marks			SEE Hrs.
					L	T	P	S	Total		CIE	SEE	Total	
1	ASC	25MA301	**Probability and Laplace Transforms**	MA	42	-	-	48	90	3	50	50	100	3
2	PCC	25ME302	**Engineering Thermodynamics**	ME	42	-	-	48	90	3	50	50	100	3
3	IPCC	25MEI303	**Strength of Materials**	ME	42	-	28	50	120	4	50	50	100	3
4	PCC	25ME304	**Materials Science and Metallurgy**	ME	42	-	-	48	90	3	50	50	100	3
5	PCC	25ME305	**Machine Tools and Manufacturing Process.**	ME	42	-	-	48	90	3	50	50	100	3
6	PCC	25AI306	**Foundations of Artificial Intelligence**	Any	28	-	-	32	60	2	50	50	100	3
7	PCC	25MEL307	**Machining Practice Laboratory**	ME	-	-	28	2	30	1	50	50	100	3
8	AECL	25MEAEL308	**Computer Aided Machine Drawing**	ME	14	-	28	18	60	2	50	50	100	3
9	HSC	25IK309	**Indian Knowledge System**	Any	14	-	-	16	30	1	100	-	100	-
				Total	**266**	**-**	**84**	**310**	**660**	**22**	**500**	**400**	**900**	**-**
For lateral entry students only														
1	HSC	25HV310	**Universal Human Values**	Any	14	-	-	16	30	1	100	-	100	-

L: Lecture, T: Tutorial, P: Practical / Drawing, S: Student Academic Activity. C – Credits, CIE: Continuous Internal Evaluation, SEE: Semester End Examination

Department: MATHEMATICS			Semester:	III
Course: PROBABILITY AND LAPLACE TRANSFORMS				
Course Code:	25ME301	L – T – P - S:	42-0-0-48=90	(In Hrs)
			Credits:	3

Sl. No	Course Objectives
This course will enable the students to :	
1	Understand the concept of fundamental statistics correlation and regression, fitting of a curve, Probability and Laplace transform.
2	Learn the mathematical techniques and methods used for solving Engineering problems involving probability, partial differential equation and Laplace transform.
3	Acquire the knowledge of complex analysis, statistics and probability distribution in Engineering problems.
4	Study the MATLAB tool for solving, visualization and interpreting Mathematical models arising in Engineering application.

Module	Description	Hrs
I	**Statistics:** Central moments, mean, variance, coefficients of skewness and kurtosis in terms of moments. Curve fitting by the method of least squares- Fitting of the straight line, second degree parabola and exponential form of the curve $= ab^x$ (All results without proof) – Problems. Definition of Correlation-Karl Pearson's coefficient of correlation–problems, Regression lines (All results without proof)–Problems. Using modern tool to solve and visualize statistical analysis problems. [Text I: 25.7, 25.12 to 25.14, 24.5, 24.6,24.8 Text III: MATLAB]	**8**
II	**Laplace Transforms:** Basic definition, Laplace transforms of elementary functions, Properties of Laplace transforms, Laplace transforms of Periodic function, Unit step function (All results without proof)-Problems only. **Inverse Laplace transforms:** Basic definition, Evaluation of inverse Laplace transforms by standard methods. - Problems only. Applications: Solutions of second order linear differential equations using Laplace transforms method. Solving and visualizing Laplace transforms using modern tools. [Text I: 21.3 to 21.5, 21.17, Text III: MATLAB]	**8**
III	**Probability Distributions:** Review of basic probability theory**.** Random variables (Discrete and Continuous), Probability of mass/density functions. Binomial distribution, Poisson's distribution, exponential distribution and Normal distribution (without derivations) and problems. Solve exact probabilities, plot distributions using modern digital tools. [Text II: 2.5 Text III: MATLAB]	**8**
IV	**Numerical Methods for Partial Differential Equations:** Numerical solutions to partial differential equations – Finite difference approximation to derivatives, solution of Laplace equation in two-dimension, heat and wave equations in one dimension (explicit methods). Using modern tool to solve numerical methods for partial differential equations. [Text I: 33.1, 33.3, 33.5, 33.8, 33.10. Text III: MATLAB]	**9**
V	**Complex Analysis:** Complex function, analytic function, Cauchy-Riemann equations, (All results without proof), harmonic functions. Construction of analytic function, Milne -Thomson method –problems. [Text I: 20.1, 20.3, to 20.5, 20.13. 20.17, 20.18]	**9**

Text books:

Sl No	Text Book title	Author	Volume and Year of Edition
I	Higher Engineering Mathematics	B.S. Grewal	44th Edition Khanna Publications, 2021.ISBN:9788193328491
II	Introduction to Probability Models	Sheldon M. Ross	Elsevier, 9th Edition, 2007
III	MATLAB Programming for Engineers	Stephen J Chapman	7th Edition, Cengage, 2024. ISBN: 9798214001531

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Probability, Random variables and Stochastic Process	Athanosios Papoulis & S. Unnikrishna Pillai	Tata McGraw Hill, 4th edition, 2002
2	Higher Engineering Mathematics	B.V.Ramana, Tata McGraw-Hill	11th Edition, Tata McGraw-Hill, 2017, ISBN:9780070634190
3	Statistics Random Processes & Queuing Theory	B.Prabha, P.S. Arunachalum,R. Sujatha	Scitech Publications,2003
4	MATLAB Programming for Engineers	Stephen J Chapman	7th Edition,Cengage,2024, ISBN:9798214001531

E Resource's Links: (for self-study)

1. https://nptel.ac.in/courses/111/105/111105035/
2. https://nptel.ac.in/courses/111/107/111107119/
3. https://nptel.ac.in/courses/111105134/
4. https://nptel.ac.in/courses/111108144/

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	**Understand** the concepts of Curve Fitting, Correlation, Regression, probability distribution. Laplace transform and Complex analysis.
CO2	**Apply** the fundamental knowledge of probability distribution, Correlation, Regression, Laplace transform Complex analysis and Numerical method partial differential equation.
CO3	**Analyze** the complex engineering problems using appropriate Mathematical techniques from Laplace transform, probability distribution, Numerical methods and Curve Fitting, Correlation, Regression,
CO4	Use modern tools to **solve/visualize** engineering problems
CO5	**Enhance** the comprehensive **understanding** of Probability distributions, complex analysis, partial differential equation, Laplace transform, arising in real word situations.

Course Articulation Matrix

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2											
CO2	3	2	1										
CO3	3	3	2								1		
CO4					2						1		
CO5	3	2	1								1		

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks						
Sl. No	**Components**		**Duration (min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory- Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL1	-	25	5	4
		ABL2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module

Activity Based Learning(ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program(1 week/online or offline), ……Etc

Semester End Examination(SEE) : Max 100 Marks, Duration: 3 Hrs

The question paper shall contain **10 full questions (20 marks each),** with **2 questions from each of the 5 Modules,** ensuring **100% syllabus coverage.** Students shall answer **any 5 full questions** with **1:1 choice**. Each question may have a **maximum of 4 sub-divisions**.

Minimum passing Marks : 40% of Max Marks

Department: Mechanical Engineering			Semester:	III
Course: ENGINEERING THERMODYNAMICS				
Course Code:	25ME302	L – T – P - S:	42-0-0-48=90	(In Hrs)
			Credits:	3

Sl. No	Course Objectives
This course will enable the students to :	
1	**Understand** the fundamental concepts of thermodynamics, thermodynamic systems, properties, processes, temperature scales and temperature measurement used in engineering applications.
2	**Understand** the thermodynamic behavior of gases and pure substances through equations of state, property relations, thermodynamic charts, and steam tables.
3	**Understand** and **develop** the ability to analyze heat and work interactions and evaluate energy transfer processes in engineering systems.
4	**Apply** the principles of the First Law of Thermodynamics to energy analysis of closed systems, open systems, and steady-flow engineering devices.
5	**Remember** and **Apply** the concepts of the Second Law of Thermodynamics and entropy for assessing the feasibility, efficiency, and irreversibility of thermodynamic processes and cycles.

Module	Description	Hrs
I	**Basic Concepts:** Thermodynamics: definition and applications, Microscopic and Macroscopic approaches, Open & Closed systems, system, boundary and control surface, examples, Properties, State, Processes and Cycles, Quasi-static process, Equilibrium of systems, Equality of temperature, Zeroth law of thermodynamics, Temperature scales, *Simple Numerical problems on temperature scales*. **Text 1: Chapter 1: 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 1.10, 1.11, 1.12, 1.13, 1.14 & Chapter 2: 2.1, 2.2, 2.3, 2.4.** **Ideal and perfect gases**: Ideal gas laws, differences between perfect, ideal and real gases, equation of state, Universal and characteristic gas constants evaluation of properties of perfect and ideal gases. Real Gases: Introduction. Vander Waal's Equation of state, Vander Waal's constants in terms of critical properties, law of corresponding states, compressibility factor; compressibility chart. *Simple Numerical problems.* **Text 1: Chapter 4: 4.1, 4.2, 4.3, 4.4, 4.5, 4.6, 4.7, 4.8, 4.9, 4.10.**	**9**
II	**Pure substance:** Definition of a pure substance, phases of a substance, triple point and critical points, sub-cooled liquid, saturated liquid, dryness fraction, vapor pressure, two-phase mixture of liquid and vapor, saturated vapor and superheated vapor states of a pure substance with water as example. Representation of pure substance properties on P-T, T-h, T-S and P-V diagrams, Calorimeters, Steam tables and its use, *Simple Numerical*	**8**

	problems. **Text 1: Chapter 12: 12.1, 12.2, 12.3, 12.4, 12.5, 12.6, 12.7, 12.9, 12.11, 12.12, 12.13, 12.14.**	
III	**Work and heat:** Thermodynamic definition of work; examples, sign convention, Displacement work; Expressions for displacement work-plotting on p-v diagrams, Other forms of work; electric work, shaft work, paddle wheel work, spring work. Free expansion with zero work transfer. Heat: definition, units and sign convention, Comparison of heat and work, *Simple Numerical problems.* **Text 1: Chapter 3: 3.1, 3.2, 3.3, 3.4, 3.5, 3.6.**	**8**
IV	**First law of thermodynamics:** Joule's experiments, equivalence of heat and work, Statement of the First law of thermodynamics, Extension of the First law to non -cyclic processes, Energy, Energy as a property, Modes of energy, Specific heats, Internal energy and enthalpy of ideal gas, PMMK-1, Principle of Conservation of mass, Energy balance for steady flow system, Some steady flow engineering devices, *Simple Numerical problems*. **Text 1: Chapter 5: 5.1, 5.2, 5.3, 5.4, 5.5, 5.6, 5.7 & Chapter 6: 6.1, 6.2, 6.3, 6.4, 6.5, 6.6.**	**8**
V	**Second law of Thermodynamics:** Limitations of First Law, Statements of second law, Thermal Energy reservoirs, Equivalence of Kelvin Plank and Clausius Statements, Heat Engines, Energy Conversion efficiencies, Refrigerators and Heat Pumps, Coefficient of performance, Perpetual motion machines, Reversible process, Factors that make a process irreversible, Carnot cycle, Carnot's Heat engine (reversible heat engines), Carnot Principle, Thermodynamic temperature scale, *Simple Numerical problems*. **Text 1: Chapter 7: 7.1, 7.2, 7.3, 7.4, 7.5, 7.6, 7.7, 7.8, 7.9, 7.10.** **Entropy:** Clausius inequality; statement, proof, definition, a property, Entropy, Increase of Entropy principle, Entropy generation, entropy as a quantitative test for irreversibility, isentropic process, property diagrams involving entropy, Tds relation, entropy change for liquids, solids, and gases (Ideal),Entropy balance, *Simple Numerical problems.* **Text 1: Chapter 7: 7.11 and Chapter 8: 8.1, 8.2, 8.3, 8.4, 8.5, 8.6, 8.7.**	**9**

Textbooks:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Engineering Thermodynamics (Principals and Practices)	Dr. D S Kumar	S.K. Kataria & Sons; Reprint 2013, 6th edition (2012).
2	Thermodynamics An engineering approach	Yunus.A.Cenegel, Michael .A .Boles	Tata McGraw hill, 2002.

Reference Books:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Engineering Thermodynamics	J.B.Jones and G.A.Hawkins,	John Wiley and Sons Ltd; 2nd edition (May 7, 1986)
2	Thermal Engineering	R.K. Rajput,	Laxmi Publications; Tenth edition (2018)
3	A Course in Thermal Engineering	Domakundawar	Dhanpat Rai& Co. (P) Limited (2016)

E Resource's Links: (for self-study)

1. www.youtube.com/watch?v=-EyzhMXIVQw&list=PLbRMhDVUMngeQ_BsUnF5PMBAbCYzYzUNx
2. www.youtube.com/watch?v=9GMBpZZtjXM&list=PLD8E646BAB3366BC8
3. www.youtube.com/watch?v=z2eUCNHkxEA&list=PLbMVogVj5nJRL8ZVcaoXG_Bbw2G3NX5N9
4. https://nptel.ac.in/courses/112105123

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	**Apply** thermodynamic principles to identify and characterize systems, properties, processes, and gas behavior, and solve numerical problems related to temperature scales and gas laws.
CO2	**Explain** and **determine** the properties and states of pure substances using thermodynamic charts, steam tables, and phase diagrams.
CO3	**Apply** and **evaluate** heat and work transfer in thermodynamic processes and calculate energy interactions in engineering applications.
CO4	**Analyze** energy conservation in closed and open systems using the First Law of Thermodynamics and perform energy balance calculations for steady-flow devices.
CO5	**Apply** the Second Law of Thermodynamics and entropy principles to determine the efficiency, coefficient of performance, and irreversibility of thermodynamic cycles and energy conversion devices.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2			1						1		
CO2	3	2		1	2						1		
CO3	3	3		1	1						1		
CO4	3	3	1	2	1		1				1		
CO5	3	3	2	2	1		2				2		

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks						
Sl. No	**Components**		**Duration (min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory- Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL1	-	25	5	4
		ABL2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module

Activity Based Learning(ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program(1 week/online or offline), ……Etc

Semester End Examination(SEE) : Max 100 Marks, Duration: 3 Hrs

The question paper shall contain **10 full questions (20 marks each),** with **2 questions from each of the 5 Modules,** ensuring **100% syllabus coverage.** Students shall answer **any 5 full questions** with **1:1 choice**. Each question may have a **maximum of 4 sub-divisions**.

Minimum passing Marks : 40% of Max Marks

Department: Mechanical Engineering			Semester:	III
Course: STRENGTH OF MATERIALS				
Course Code:	25MEI303	L – T – P - S:	42-0-28-50=120 (In Hrs)	
			Credits:	4

Sl. No	Course Objectives
This course will enable the students to :	
1	**Explain** the basic concepts of stress, strain, deflection, principal stresses, and principal strains.
2	**Analyze** beams subjected to various loading conditions.
3	**Evaluate** the bending moment, strength, and stiffness of beams.
4	**Analyze** the stress distribution in circular shafts subjected to torsion.
5	**Analyze** the stress distribution in thick and thin cylinders subjected to internal pressure.

Module	Description	Hrs
I	**Simple Stress and Strain**: Introduction, Properties of Materials, stress, strain, Hook's law, Poisson's Ratio, Stress-Strain diagram for ferrous and non-ferrous materials, Principles of super position, Elongation of tapering bars of circular and rectangular cross sections. Elongation due to self-weight. **Volumetric strain:** expression for volumetric strain, elastic constants, relationship among Elastic constants, thermal stresses including compound bars. Strain energy – Resilience – Gradual, sudden, impact and shock loadings. State of Stress in two dimensions — Stresses on inclined planes, plane stress and plane strains T-1: 8.1 to 8.21 R-1:1.2 to 1.14	**9**
II	**Bending moment and shear force in beams:** Introduction, types of beam loading and supports, shearing force in beams, bending moment, sign convention, relationship between loading shear force and bending moment, SFD, BMD for cantilever beams, simply supported beams and overhanging Beams considering point loads, UDL, UVL, Couple. Point of contra flexure. T-2; 6.1 to 6.18 R-1: 4.1 to 4.21	**9**
III	**Bending stress and shear stress in beams:** Introduction, Bending stress in beams, assumptions in simple bending theory and pure bending, modulus of rupture, section modulus, flexural rigidity, beam of uniform strength, expression for horizontal shear stress in beams, shear stress diagram for rectangular, symmetrical I and T section. Determinate and	**8**

	indeterminate structures (Fletched beams not included) **Principal stresses and strains**: Principal planes, principal stresses and strains, biaxial state of stress combined with shear, concept of Mohr's circle diagram. T-2; 7.1 to 7.11, T-2; 3.1 to 3.6, R-1: 5.1 to 5.8, R-1: 2.1 to 2.8	
IV	**Deflection of beams:** Derivation of the relations between curvature, slope, deflection and moment. Assumptions, methods of determining slope and deflections. Determination of deflection of a simply supported beam Subjected to concentrated load at centre, subjected to UDL, cantilever beam subjected to point load at free end and also UDL. conjugate beam method for computation of slope and deflection of determinant beams. **Elastic stability of columns:** Introduction, effective length, slenderness ratio, short and long columns, radius of gyration, buckling load, Failure of columns, assumptions derivations of Euler's theory, Rankine's formula. Limitations of Euler's theory, eccentric loading, secant formula T-2; 12.1 to 12.11, T-2: 19.1 to 19.11 R-1: 8.1 to 8.10, R-1:16.1 to 16.8, and 11	**8**
V	**Torsion of circular shafts:** Introduction, pure torsion equations of circular shafts, strength and stiffness, torsional rigidity, torsional flexibility, polar modulus, power transmitted by solid and hollow circular sections. Power transmitted by shafts **Thin and Thick cylinders**: thin and thick cylinders subjected to pressure, change in length, diameter, volume, Lame's equation (compound cylinders not included). T-2; 16.1 to 16.14, 17, 18 R-1: 13.1 to 13.10, R-1 10	**8**

Lab Content

Sl. No	Experiment Description
1	Conduction of tensile test of metallic and nonmetallic specimen Using a universal testing machine.
2	Conduction of shear test of metallic and nonmetallic specimen using a Universal testing machine.
3	Conduction of impact test(Izod and Charpy) test on mild steel specimen.
4	Experiments on wear studies using ferrous, nonferrous & composite materials for Different parameters.
5	Brinell, Rockwell and Vicker's hardness test.
6	Torsion test
7	Fatigue test.
8	Buckling of long column test,

9	Conduction of compression test of metallic and nonmetallic specimen
10	Conduction of bending of metallic and nonmetallic specimen using a Universal testing machine.

Textbooks:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Strength of Materials	S.S.Bhavikatti	Vikas Publications Pvt.Ltd., New Delhi,2016.
2	Strength of Materials	R K Bansal	Laksmi Publications, New Delhi,2018.
3	Strength of materials	R S Kurmi	*S* Chand *Publications* Edition- 2008

Reference Books:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Mechanics of Solids	R K Rajput	*S* Chand, Classic Edition- 1 July 2016
2	Strength of Materials	B.C.Punmia, AshokJain	Lakshmi Publications,2010.
3	Strength of materials	Ramamrutham	Dhanpat Rai publishers, New Delhi,2014.
4	Strength of materials	S Timoshenko	D Van Nostrand company, 1930,1940

E Resource's Links: (for self-study)

1. https://study.madeeasy.in/subjects/what-is-strength-of-material
2. http://www.digimat.in/nptel/courses/video/112107147/L34.html
3. https://youtu.be/uA_HqCGo8Pg?si=s-lwXL3GENtiTE4k
4. https://youtu.be/GkFgysZC4Vc?si=9u43Y3UTL_CxxNHb
5. https://youtu.be/iNG4bLMyeFA?si=1egsfxOWiabbmVeI
6. https://youtu.be/EyIEenmUUfU?si=RtUiP_IVtyFrXo3V
7. https://youtu.be/iJTHPgYAqvQ?si=H8dXPC1VAKVtKgnW

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	**Understand** the concepts of stresses, strains, deflections, principal Stresses and strains.
CO2	**Analyze** and **apply** the knowledge of Beams under various loading and support conditions.
CO3	**Analyze** the stress distribution in the shaft of circular cross section,Strength deflection and rigidity criteria.
CO4	**Understand** the Analysis of thick and thin cylinders subjected to internal Pressure
CO5	**Analyze** structural loading of various deformable bodies subjected to axial bending, torsion and shear conditions

Course Articulation Matrix

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	1				1	1		2		
CO2	2	2	1	1				1	1		1		
CO3	3	3	3	1				1	1		1		
CO4	3	2	2	3				1	1		1		
CO5	3	2	2	3				1	1		1		

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks						
Sl.No	**Components**		**Duration (min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory-Internal Assessment	IA-1	90	50	15	12
		IA-2	90	50	15	
2	Periodic Lab Assessment		-	30	10	4
3	Lab CIE		-	20	5	2
4	One Open Ended Experiment/ Project		-	10	5	2
	Total CIE			**160**	**50**	**20**

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module
Rubrics for Periodic Lab Assessment: Attendance (5M), Lab record (10 M), Conduction of Experiments(10M) and Viva(5M)-Total 30M
Lab CIE: Write-up(5M), Conduction/Execution/Demonstration(10M) and Viva(5M)-Total 20M

Semester End Examination(SEE) : **Max 100 Marks, Duration: 3 Hrs**
The question paper shall contain **10 full questions (20 marks each),** with **2 questions from each of the 5 Modules**, ensuring **100% syllabus coverage**. Students shall answer **any 5 full questions** with **1:1 choice.** Each question may have a **maximum of 4 sub-divisions.**
Minimum passing Marks : 40% of Max Marks

Department: Mechanical Engineering			Semester:	III
Course: MATERIALS SCIENCE AND METALLURGY				
Course Code:	25ME304	L – T – P - S:	42-0-0-48=90 (In Hrs)	
			Credits:	3

Sl. No	Course Objectives
This course will enable the students to :	
1	**Understand** the relationships among the structure, properties, processing, and applications of various engineering materials.
2	**Analyze** the various modes of failure in mechanical engineering materials and their influence on engineering performance.
3	**Explain** the phase transformations in metals and the micro structural and property changes in iron–carbon alloys.
4	**Apply** the knowledge of composite materials, their processing methods, and other advanced engineering materials for suitable engineering applications.

Module	Description	Hrs
I	**Mechanical Behavior:** Stress-strain diagrams to show ductile behaviour of materials,. Mechanical properties in plastic range yield strength, offset yield strength, ductility, ultimate tensile strength, toughness and yield point phenomena. Nano-structured Materials – Introduction and Application. **Text 1: Ch. 5, 16** **Text 2: Ch. 6, 22**	**9**
II	**Fracture, fatigue & creep:** Introduction, Fracture, Types of fracture, Ductile & Brittle fracture, Fatigue, fatigue Test, S-N curve, factors affecting fatigue life, fatigue protection methods, Creep, creep Test, creep curve, factors affecting creep. **Solidification:** Mechanism of solidification, crystal growth, Cast metal structures. **Text 1: Ch. 2, 6, 7, 8** **Text 2: Ch. 8, 10**	**9**
III	**Phase diagram:** Solid solutions, substitution, and interstitial solid solutions, Hume Rothary rules, Phase Rule - Gibbs phase rule, Types of phase diagram, construction and Interpretation of Phase diagrams, Binary Phase diagrams lever rule. Numerical on phase diagrams. **Iron - Carbon system:** Iron - carbon equilibrium diagram description of phases, Solidification of steels and cast irons and invariant reactions. **Text 1: Ch. 3, 4, 5** **Text 2: Ch. 9, 10, 11**	**8**
IV	**Heat treatment of metals:** Time-Temperature-Transformation (T-T-T) diagrams, continuous cooling Transformation diagrams, Annealing and its types, Normalizing, Hardening, Tempering, Martempering, Austempering,	**8**

	Hardenability, Vacuum Heat Treatment, Powder Bed Fusion. **Introduction to Composite Materials:** Definition, Classification of composites, advantages, limitations and application of composites. **Text 1: Ch. 10, 11, 12** **Text 2: Ch. 11, 13, 16**	
V	**Manufacturing of Composites:** Lay-up and curing - open and closed mould processing, Hand lay-Up techniques, spray Lay-up process, filament winding, Pultrusion, compression moulding, stir casting, Sheet Moulding compound (SMC) process. **Smart Materials:** Introduction, Classification and application of smart materials. **Text 1: Ch. 12, 13, 15** **Text 2: Ch. 16, 17, 20**	**8**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Materials science and metallurgy for engineers	Dr. V.D. Kodigere	Everest Publishing House, Pune, 38th Edition, 2016.
2	Material science and Engineering	William D. Callister	Ed. 5, John wiley& sons, 2001

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Foundations of Materials Science and Engineering	Smith	3rd Edition McGraw Hill, 1997.
2	Structure and properties of Engineering materials	V.S.R Murthy	Tata McGraw Hill, 2003.
3	An Introduction to Metallurgy	Alan Cottrell	University Press India, Oriental Longman Pvt. Ltd, 1974.

E Resource's Links: (for self-study)

1. https://nptel.ac.in/courses/113106099
2. https://www.asminternational.org
3. https://www.tms.org
4. https://www.doitpoms.ac.uk

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	**Analyze** the mechanical behavior of engineering materials by interpreting stress–strain diagrams, distinguishing linear and nonlinear elasticity, and evaluating properties such as yield strength, ductility, toughness, and the mechanisms of plastic

	deformation (e.g., slip and twinning).
CO2	**Examine** and differentiate the failure modes of materials—including fracture, fatigue, and creep—by conducting standard mechanical tests, interpreting S-N and creep curves, and applying methods for enhancing fatigue and creep resistance in engineering applications.
CO3	I**nterpret** and **construct** phase diagrams for binary alloy systems using Gibbs phase rule and lever rule, and analyze the iron-carbon equilibrium diagram to explain phase transformations, solidification, and invariant reactions relevant to steels and cast irons.
CO4	**Design** and assess heat treatment processes for metals by applying T-T-T and continuous cooling diagrams, evaluating processes such as annealing, normalizing, hardening, tempering, and surface hardening, and relating microstructural changes to mechanical properties.
CO5	**Evaluate** manufacturing methods for composite and smart materials by comparing open and closed mold processing techniques, assessing properties, advantages, and limitations of composites and smart materials, and recommending suitable materials for specific engineering applications.

Course Articulation Matrix

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	2	1	1	1	1	1	2	1		
CO2	3	3	2	3	2	1	1	1	2	2	1		
CO3	3	2	3	2	2	1	1	1	2	2	2		
CO4	3	2	3	3	2	1	1	1	2	2	2		
CO5	3	2	3	2	2	2	2	1	2	2	2		

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks						
Sl. No	**Components**		**Duration (min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory- Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL1	-	25	5	4
		ABL2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module **Activity Based Learning(ABL):** Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program(1 week/online or offline), ……Etc
Semester End Examination(SEE) : Max 100 Marks, Duration: 3 Hrs The question paper shall contain **10 full questions (20 marks each),** with **2 questions from each of the 5 Modules,** ensuring **100% syllabus coverage.** Students shall answer **any 5 full questions** with **1:1 choice**. Each question may have a **maximum of 4 sub-divisions**. **Minimum passing Marks : 40% of Max Marks**

Department: Mechanical Engineering			Semester:	III
Course: MACHINE TOOLS AND MANUFACTURING PROCESS				
Course Code:	25ME305	L – T – P - S:	42-0-0-48=90 (In Hrs)	
			Credits:	3

Sl. No	Course Objectives
This course will enable the students to :	
1	To **develop fundamental knowledge** on metal cutting parameters, tool materials, cutting fluids and tool wear mechanisms.
2	To **understand the fundamentals** of machining processes and machine tools.
3	To **understand the need** of work holding and tool holding devices in machine tools.
4	To **apply knowledge** of basic mathematics to calculate the machining parameters for different machining processes.

Module	Description	Hrs
I	**Theory of Metal Cutting**: Single point cutting tool nomenclature, tool geometry, Chip formation, Orthogonal and Oblique cutting, Shear angle relationship (only formula), tool wear , tool failure, tool life, effects of cutting parameters on tool life, tool failure criteria, Taylor's tool life equation, numerical. Cutting tool materials: Properties, types of cutting tool materials – HSS, coated carbide tools, carbide tools, ceramics, CBN, Cutting fluids: properties, types and selection of cutting fluids Heat generation in metal cutting: Factors affecting heat generation, heat affected zones. **Text-1:1.1,1.2,2.1-2.6,2.8-2.10,2.12,2.14**	**9**
II	**Lathe:** Classification of lathe, constructional features of capstan and turret lathe, lathe operations, tool layout, tool and work holding devices, machining time calculation. **Drilling Machines:** Classification of drilling machines, constructional features of Bench drilling and Radial drilling machine, tool and work holding devices, types of drill bits, nomenclature of twist drill, machining time calculation. **Text-1:3.1,3.2,4.1-4.10,8.1-8.6**	**9**
III	**Shaping and Planing Machines**: Classification, constructional features of horizontal shaper, constructional features of double house planner, Operations, tool and work holding devices, machining time calculations, **Broaching Machines:** Classification, principle of broaching, nomenclature of broach, application of broaching machine. **Text-1:6.1-6.5**	**8**
IV	**Milling Machines**: Classification of milling machines. Constructional features of Column and knee milling machine, milling cutters- types. Cutter nomenclature. Milling operations, machining time calculation. Indexing: Indexing mechanism, Simple, Angular types of indexing, numerical. **Text-1:7.1-7.8**	**8**

V	**Grinding Machines:** Classification of grinding machines, constructional features of grinding machines, grinding wheels-types, grinding wheel nomenclature, applications. **Text-1:9.1-9.5**	**8**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Manufacturing Technology Metal cutting and Machine tools.	P.N. Rao	Tata McGraw-Hill publications,2018.

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Workshop Technology	Hazra Choudhry	Vol-II, Media Promoters & Publishers Pvt Ltd,2004
2	Production Technology.	R.K. Jain	Khanna Publications,2003
3	Production Technology.	HMT	TATA McGraw Hill, 2001

E Resource's Links: (for self-study)

1. https://www.youtube.com/playlist?list=PLSGws_74K018JY-1Rylj0cm4yppa1h54r
2. https://www.slideshare.net/slideshow/theory-of-metal-cutting-78791391/78791391

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	**Identify** cutting tool geometry and **explain** the principles of metal cutting.
CO2	**Interpret** the specifications of various machine tools.
CO3	**Explain** the basic working principles of machine tools.
CO4	**Perform** basic machining operations on various machine tools.
CO5	**Analyze** the process parameters of various machine tools.

Course Articulation Matrix

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1		3					2					
CO2													
CO3	3			3	2						1		
CO4	1		2										
CO5					1			2			3		

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks						
Sl. No	**Components**		**Duration (min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory- Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL1	-	25	5	4
		ABL2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module

Activity Based Learning(ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program(1 week/online or offline), ……Etc

Semester End Examination(SEE) : Max 100 Marks, Duration: 3 Hrs

The question paper shall contain **10 full questions (20 marks each),** with **2 questions from each of the 5 Modules,** ensuring **100% syllabus coverage.** Students shall answer **any 5 full questions** with **1:1 choice**. Each question may have a **maximum of 4 sub-divisions**.

Minimum passing Marks : 40% of Max Marks

Department: Mechanical Engineering			Semester:	III
Course: FOUNDATIONS OF ARTIFICIAL INTELLIGENCE				
Course Code:	25AI306	L – T – P - S:	28-0-0-32=60 (In Hrs)	
			Credits:	2

Sl. No	Course Objectives
This course will enable the students to :	
1	**Introduce** basic concepts of Artificial Intelligence.
2	**Learn** the techniques in Artificial Intelligence.
3	**Understand** the Engineering applications and basics of Robotics.
4	**Realize** ethical approaches in Artificial Intelligence.

Module	Description	Hrs
I	**Introduction to Artificial Intelligence:** Definition of Artificial Intelligence, History of Artificial Intelligence, Types of Artificial Intelligence, Augmented Intelligence and Cognitive Computing, Machine Learning and Deep Learning. Text Book 1: 1.1-1.5	**5**
II	**Artificial Intelligence Technologies:** Techniques in AI, Machine Learning Model, Types of Machine Learning Algorithms, Regression Analysis in Machine Learning, Working of Regression Analysis, Model Evaluation Metrics, Types of Regression Text Book 1: 2.1-2.3	**6**
III	**Classification Techniques**: K-Nearest Algorithm, Decision Trees, Random Forests **Clustering Techniques**: K-Means Algorithm, Applications of Clustering in Real-World Scenarios, Evaluation Metrics for Clustering, Working of K-Means Algorithm, Pros and Cons of K-Means Algorithm, Density-Based Spatial Clustering of Applications with Noise (Dbscan) Text Book 1: 2.4-2.5	**6**
IV	**AI to all:** AI and Society, AI and Privacy, AI in Healthcare, AI in Finance, AI in Marketing, AI and Automation, AI-Powered Communication, AI in Customer Service, Future Trends in AI, AI for Personal Growth. Text Book 2: 10-21 **Robotics - An Application of AI:** Drones Using AI, Future of AI, No Code AI, Low Code AI Text Book 1: 1.7, 1.8, 1.9, 1.10	**5**
V	**Ethical Decision-Making:** Introduction, Ethical Theories, Values, Ethics in Practice, Implementing Ethical Reasoning**.** **Taking Responsibility:** Introduction, Responsible Research and	**6**

	Innovation, The ART of AI: Accountability, Responsibility, Transparency, Design for Values. Text Book 3: 3, 4, 5-5.1,5.2, 5.3	

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Artificial Intelligence: Beyond Classical AI,	Reema Thareja	Pearson Education, 2023 ISBN-10 : 9356069328
2	AI for Everyone	Kai L Thornton	eBooklt.com, 2025 ISBN-13 : 978-1456664169
3	Responsible Artificial Intelligence	Virginia Dignum	Springer-Nature New York Inc , 2019. ISBN-10: 3030303705

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Artificial Intelligence Structure and Strategies for Complex Problem Solving	George F Lugar	Pearson Education, 2011, 5th Edition, ISBN-10 : 8131723275
2	Artificial Intelligence A Modern Approach	Stuart Russell & Peter Norvig	Pearson Education, 2010 3rd Edition, ISBN-10 : 9789332543515
3	Artificial Intelligence	Saroj Kaushik	Cengage learning, 2014 1st Edition, ISBN-10 : 8131510999

E Resource's Links: (for self-study)

1. https://nptel.ac.in/courses/112103280
2. https://www.youtube.com/playlist?list=PLwdnzlV3ogoXaceHrrFVZCJKbm_laSHcH
3. https://nptel.ac.in/courses/106106472
4. https://www.youtube.com/watch?v=mmrlkO4vpts

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	**Outline** the types of Artificial Intelligence.(L2)
CO2	**Identify** the different Regression models. (L3)
CO3	**Analyze** the classification techniques in Artificial Intelligence.(L4)
CO4	**Extend** the applications of Artificial Intelligence. (L2)
CO5	**Relate** ethical decision making.(L2)

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2											
CO2	2	2	2	1									
CO3	2	2	2	1									
CO4	2	2	2	1									
CO5	2	1	1	1		2	3						

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks

Sl. No	Components		Duration (min)	Marks Conducted for	Marks reduced to	40% Min Marks in each component to write SEE
1	Theory- Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL1	-	25	5	4
		ABL2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module

Activity Based Learning(ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program(1 week/online or offline), ……Etc

Semester End Examination(SEE) : Max 100 Marks, Duration: 3 Hrs

The question paper shall contain **10 full questions (20 marks each),** with **2 questions from each of the 5 Modules,** ensuring **100% syllabus coverage.** Students shall answer **any 5 full questions** with **1:1 choice**. Each question may have a **maximum of 4 sub-divisions**.

Minimum passing Marks : 40% of Max Marks

Department: Mechanical Engineering			Semester:	III
Course: MACHINING PRACTICE LABORATORY (Professional Core Laboratory - 1)				
Course Code:	25MEL307	L – T – P - S:	0-0-28-2=30	(In Hrs)
			Credits:	1

Sl. No	Course Objectives
This course will enable the students to :	
1	**Explain** the working principles and safety practices of machine shop equipment such as lathe, drilling, milling, and slotting machines.
2	**Apply** appropriate workpiece setup and cutting tool selection techniques to perform machining operations using machine tools.
3	**Perform** turning, drilling, milling, and slotting operations to manufacture components to specified dimensions.
4	**Analyze** machine tool parameters such as cutting speed, feed, and depth of cut and their influence on machining quality and productivity.

	Description	Hrs
PART A	Turning operations: Preparing a model involving the turning operations like Facing, Drilling, Plain Turning, Step Turning, Taper turning by swiveling of compound rest method, Thread cutting (V-Thread), Knurling.	**10**
PART B	Milling operations: Prepare the spur gear from Aluminium disc involving operations like forward milling, indexing, and cutting at least 3 teeth's of a gear. Shaping operation: Prepare the given V slot or Rectangular slot in a given iron block using shaping machine.	**10**
Question paper Pattern: The student has to Perform Part A, turning operations to convert given raw material to specified dimensions of specimen in lathe individually (25 Marks). The pair of students has to operate milling or shaping machine to complete the given specimen (15 Marks). Viva for 10 marks.		

Textbooks:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Manufacturing Technology	P N Rao	4th Edition, Published: July 24, 2018, Mcgrew hill publication
2	Manufacturing Science and Technology Manufacturing Processes and Machine Tools	K. Vara Prasada Rao	2nd edition, New Age International Pvt Ltd Publishers, 2009

Reference Books:

Sl No	Textbook title	Author	Volume and Year of Edition
1	All About Machine Tools	H. Gerling	New Age International Private Limited, 2006
2	Fundamentals of Machining and Machine Tools	R K Singal	Dreamtech publishers, 2020

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	**Explain** the working principles, constructional features, and safety practices associated with lathe, drilling, milling, and slotting machines.
CO2	**Perform** basic machining operations such as turning, facing, drilling, reaming, milling, and slotting using conventional machine tools.
CO3	**Manufacture** simple components to specified dimensions and tolerances by applying appropriate machining processes.
CO4	**Inspect** machined components and **evaluate** the effect of machining parameters on dimensional accuracy and surface finish.
CO5	**Determine** and **apply** suitable cutting speed, feed, and depth of cut to achieve the desired machining quality and productivity.

Course Articulation Matrix

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	-	-	1	2	3	1	1	-	1	1	-
CO2	3	2	-	1	1	-	2	-	-	-	1	1	-
CO3	3	3	2	2	1	-	2	-	-	-	1	1	-
CO4	2	3	-	3	1	-	1	1	1	-	-	1	-
CO5	3	3	2	2	1	-	2	-	-	-	1	1	-

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks			
Sl.No	**Components**	**Marks Conducted for**	**40% Min Marks in each component to write SEE**
1	Periodic Lab Assessment	30	12
2	Lab CIE	20	8
	Total CIE	50	20
Rubrics for Periodic Lab Assessment: Attendance (5M), Lab record (10M), Conduction of Experiments(10M) and Viva((5M)-Total 30M Lab CIE: Write-up(5M), Conduction/Execution/Demonstration(10M) and Viva(5M)-Total 20M			

Semester End Examination(SEE) : Max 50 Marks, Duration: 3 Hrs
Rubrics for Lab Examination: Write-up(8M),
Conduction/Execution/Demonstration(35M) and Viva(7M)-Total 50M.
Minimum passing Marks : 40% of Max Marks

Department: Mechanical Engineering			**Semester:**	**III**
Course: COMPUTER AIDED MACHINE DRAWING (Ability Enhancement Laboratory – 1)				
Course Code:	**25MEAEL308**	**L – T – P - S:**	**14-0-28-18 (In Hrs)**	
			Credits:	**2**

Sl. No	**Course Objectives**
This course will enable the students to :	
1	**Understand** the CAD software, graphical user interface, sketching tools, and 3D modelling techniques used in engineering drawing.
2	**Develop** the ability to create orthographic projections, sectional views, and detailed drawings of machine components following BIS standards.
3	**Impart** knowledge of standard machine elements such as riveted joints, bolts, nuts, washers, and their assembly drawings.
4	**Prepare** and **interpret** assembly drawings of common mechanical components using CAD-based drafting techniques.

Module	**Description**	**Hrs**
	PART A	
I	**Introduction to CAD software and 3D modelling from Isometric views:** Review of graphic user interface of the software and review of basic sketching commands and navigational commands. **Orthographic views and Sectional views of machine parts:** Conversion of isometric views of simple machine parts into 3D model and use it for the preparation of draft sheet by generation of orthographic views by projections of simple machine parts without sectionand sectional views. (Bureau of Indian Standards conventions are to be followed for the drawings) **Text 1: Chapter 2: 2.1, 2.2, 2.3, 2.4, 2.5, 2.6 and Chapter 3:3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 3.10, 3.11, 3.12, 3.13,**	**6+4**
	PART B	
II	**Fasteners:** Hexagonal headed bolt and nut with washer (assembly), square headed bolt and nut with washer (assembly) simple assembly using stud bolts with nut. **Text 1: Chapter 4: 4.9, 4.10, 4.11, 4.12, 4.13.** **Riveted Joints: S**ingle and double riveted lap joints, butt joints with single/double cover straps (Chain and Zig-zag).**Text 1: Chapter 5: 5.1, 5.2, 5.3, 5.4, 5.4, 5.6, 5.6, 5.7, 5.8, 5.9, 5.10.**	**6**
	PART C	
III	**Assembly drawings:** (Part drawings should be given) 1. Universal Coupling. **Text 1: Chapter 8: 8.9** 2. Screw jack (Bottle type). **Text 1: Chapter 11: 11.8** 3. Plummer block (Pedestal Bearing). **Text 1: Chapter 9: 9.5** 4. Machine vice.**Text 1: Chapter 11: 11.9.**	**2+24**

Text books:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Machine drawing	K R Gopal Krishna	Subhas publications, Bangalore.
2	Machine Drawing	N D Bhat & V M Panchal.	Charotar publishing Ltd, 2014, 50th edition

Reference Books:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Machine drawing	Sri. K L Narayan, P.Kannaiah & K.Venkat Reddy,	New-Age International Publications, 2001.
2	A Primer on Computer aided Machine drawing	VTU	Published by VTU, Belgaum

E Resource's Links: (for self-study)

1. www.youtube.com/watch?v=kxRbD_gtlJI&list=PLGiGNMkNq6QsSHEfUC1ekc2Pz2gfy1OKb
2. www.youtube.com/watch?v=G_kJ_DzB34s
3. www.youtube.com/watch?v=7o7cSwbNB5M
4. www.youtube.com/watch?v=eSqHI79rDqs
5. www.youtube.com/watch?v=VT_Tm_4cqIo

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	**Apply** CAD tools and commands to create 2D sketches and 3D models of simple machine components.
CO2	**Develop** orthographic and sectional views from isometric drawings and prepare engineering drawings in accordance with BIS standards.
CO3	**Construct** detailed 2D drawings of standard riveted joints, bolts, nuts, washers, and other fastening elements.
CO4	**Develop** the 2D and 3D assembly drawings of machine elements such as universal coupling, screw jack, plummer block and machine vice using CAD software.
CO5	**Apply** the skills required for effective use of the ***Solid Edge*** software package to create, modify and manage engineering drawings and 3D models.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	1	2		3						1		
CO2	2	2	2		3		1		2		1		
CO3	2	1	2		3				1		1		
CO4	2	2	3	1	3			1	2		1		
CO5	1		2		3				1		2		

Course Assessment and Evaluation:

This course consists of experimental and project-based learning.

- **Part A:** 5 weeks of experimental learning with 5 experiments. Students shall conduct the experiments individually. CIE for Part A shall be conducted in the 7th week.
- **Part B:** 5 weeks of project-based learning. A team of maximum 4 students shall carry out a project. The project shall be completed by the 13th week; CIE for Part B may be conducted in the 14th week.

Continuous Internal Evaluation (CIE): 50 Marks				
Sl. No	**Components**	**Marks Conducted for**	**Marks Reduced to**	**40% Min Marks in each component to write SEE**
CIE for PART A(Experimental Learning)				
1	Periodic Lab Assessment	30	15	6
2	Lab CIE	20	10	4
	Total CIE for Part A	50	25	10
Rubrics for Periodic Lab Assessment: Attendance (5M), Lab record (10 M), Conduction of Experiments(10) and Viva(5M)-Total 30M **Lab CIE: Write-up(5M), Conduction/Execution/Demonstration(10M) and Viva(5M)-Total 20M**				
CIE for PART B (Project based Learning)				
1	Lab CIE Part B	50	25	10
	Total CIE(Part A and Part B)	**100**	**50**	**20**
Rubric for Lab CIE Part B: Write-up (10M), Presentation(10M),Demonstration(20M) Viva(10M)- **Total 50 Marks**				
Semester End Examination (SEE) : 50 Marks, Duration: 3 Hrs				
PART A (25 Marks): **Rubrics for Part A: Write-up (5M), Conduction/Execution/Demonstration (15M) and Viva (5M) - Total 25M.** **PART B (25 Marks):** **Rubrics for Part B: Write-up (5M), Presentation/Demonstration (15M) and Viva (5M)-Total 25M.** **Minimum passing Marks: 40% in Part A and Part B each**				

Department: Mechanical Engineering			**Semester:**	**III**
Course: INDIAN KNOWLEDGE SYSTEM				
Course Code:	**25IK308**	**L – T – P - S:**	**14-0-0-16=30**	**(In Hrs)**
			Credits:	**1**

Sl. No	Course Objectives
This course will enable the students to :	
1	**Integrate** Indian Knowledge Systems into education and the research ecosystem to promote their rejuvenation and mainstreaming.
2	**Relate** traditional Indian Knowledge Systems with contemporary knowledge systems.
3	**Apply** Indian Knowledge Systems to address contemporary societal needs and challenges.
4	**Explain**, **analyze**, and **apply** traditional knowledge in day-to-day life.

Module	Description	Hrs
I	**Introduction to Indian Knowledge Systems:** Meaning, definition and scope of IKS, civilizational continuity of Indian culture, holistic nature of knowledge in India, Gurukul system, Patashalas, Takshashila, Nalanda, overview of Indian philosophical traditions (Darshanas)	**3**
II	**Science and Technology in Ancient & medieval India** Mathematics: Baudhayana Sulba Sutra, Aryabhata, Bhaskara, concept of zero and decimal system; Medicine: Ayurveda, Charaka and Sushruta; Engineering & Technology: Iron Pillar of Delhi, water management systems, town planning, architecture, arts, Astronomy, Astrology, Crafts, Trade in India, Ancient sculptures and monuments, space programs	**3**
III	**Society, Governance and Ethics:** Concept of dharma and social order, ethical teachings in Indian texts, Chanakya's Arthashastra and governance, traditional economic and administrative systems, Anubhava mantapa- first parliament, Indian health systems: Ayurveda and other Indian medicine, Yoga	**3**
IV	**Language, Literature and Knowledge Transmission:** Sanskrit as a scientific language, different languages, Ancient literatures, Books, Indian literary, famous poets and literarists, Indian Inventions and Contributions, Awards	**2**
V	**Indian Cultural Traditions and Practices:** Festivals and cultural diversity of India**,** Traditional family and community systems, social values, folk, arts, music, dance and theatre traditions**,** industrial revolution and indigenous technologies**,** role of traditions in promoting social harmony and cultural identity	**3**

Textbooks:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Introduction to Indian Knowledge System- concepts and applications	B Mahadevan, Vinayak Rajat Bhat, Nagendra Pavana R N,	2022, PHI Learning Private Ltd, ISBN-978-93-91818-21-0.
2	Traditional Knowledge System in India,	Amit Jha	Atlantic Publishers and Distributors (P)Ltd. ISBN-13: 978-8126912230. 2009.

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Knowledge Traditions and Practices of India,	Kapil Kapoor, Avadhesh Kumar Singh,	Vol. 1, 2005, DK Print World (P) Ltd., ISBN 81-246-0334
2	https://www.iitkgp.ac.in/department/KS;jsessionid=C5042785F727F6EB46CBF432D7683B63 (Centre of Excellence for Indian Knowledge System, IIT Kharagpur)		

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	**Explain** the concept of Indian Knowledge Systems and their importance.
CO2	**Appreciate** the need and importance of protecting traditional knowledge.
CO3	**Recognize** the relevance of traditional knowledge in different domains.
CO4	**Establish** the significance of Indian Knowledge Systems in the contemporary world.
CO5	**Recognize** the relevance of traditional knowledge in Humanities and Social Sciences.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2							3				1	
CO2		1				2				2			
CO3			2										
CO4				2							1		
CO5					3		1						

Course Assessment and Evaluation:

Sl.No	Components		Duration(min)	Marks Conducted for	Min Marks To pass the Course
1	Theory- Internal Assessment	IA-1	90	50	40
		IA-2	90	50	
	Total CIE			**100**	**40**
IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module					

UNIVERSAL HUMAN VALUES

For lateral entry students only

Department: Mechanical Engineering			Semester:	III
Course: UNIVERSAL HUMAN VALUES				
Course Code:	25HV310	L – T – P - S:	14-0-0-16=30 (In Hrs)	
			Credits:	1

Sl. No	Course Objectives
This course will enable the students :	
1	This introductory course input is intended: To help the students appreciate the essential complementarily between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.
2	To facilitate the development of a Holistic perspective among students towards life and profession as well as towards happiness and prosperity based on a correct understanding of the Human reality and the rest of existence.
3	To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct, trustful and mutually fulfilling human behavior and mutually enriching interaction with Nature
4	This course is intended to provide a much needed orientation input in value education to the young enquiring minds.

Module	Description	Hrs
I	Introduction to Value Education: Understanding Value education. Self-exploration as the process for Value Education. The Basic Human Aspirations and their fulfillment. Right Understanding, Relationship and Physical Facility. Happiness and Prosperity current scenario .Method to fulfill the basic Human Aspirations.	3
II	Harmony in the Human Being: Understanding the Human being as the Co existence of the self .Distinguishing between the Needs of the Self and the Body. The Body as an Instrument of the Self. The response of the self and the body. Understanding Harmony in the Self. Harmony of the Self with the Body. Programme to ensure self regulation and health	3
III	Harmony in the Family and Society: Understanding Harmony in the Family the basic human Interaction. 'Trust' the Foundational Value in Relationship. 'Respect' – as the right evaluation other Values in Human to human relationship. Understanding Harmony in the society. Vision for the Universal Human Order.	3
IV	Harmony in the Nature (Existence): Understanding Harmony in the Nature, interconnectedness, self-regulation, mutual fulfillment among the four orders of nature. Realizing Existence as coexistence at all levels. The Holistic Perception of Harmony in Existence.	2
V	Implications of the Holistic Understanding a look at professional Ethics: Basis for universal Human Values. Definitiveness of (Ethical) Human	3

	Conduct. Professional Ethics in the light of Right Understanding. A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order. Holistic Technologies, Production Systems and Management models, Typical Case Studies. Strategies for transition towards value based life and profession	

Textbooks:

Sl No	Textbook title	Author	Volume and Year of Edition
1	A Foundation Course in Human Values and Professional Ethics	R R Gaur, R Sangal G P Bagaria	Excel Books, New Delhi,2010
2	A Foundation Course in Holistic Human Health	Sharmila Asthana, Akhilesh Shukla, T Sundara Raj Perumall	Vol 1,UHV Publications, UHV Foundation, Delhi,2023

Reference Books:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Human Values	A.N. Tripathi	New Age Intl. Publishers, New Delhi, 2004
2	Universal human values & professional ethics:	Karthik raj dasari	Vol 1 , Notion Press,2025

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	By the end of the course, students are expected to become more aware of themselves, and their surroundings family, society, nature they would become more responsible in life, and in handling problems with sustainable solutions, while keeping human relationships and human nature in mind
CO2	They would have better critical ability, also become sensitive to their commitment towards what they have understood (human values, human relationship and human society
CO3	It is hoped that they would be able to apply what they have learnt to their own self in different day to-day settings in real life, at least a beginning would be made in this direction
CO4	This is only an introductory foundational input. It would be desirable to follow it up by faculty student programs throughout their time
CO5	Leads to form strategies for transition towards Value based life and profession.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1						3	3	3	2	3			
CO2				2		3	3	3	2	3		1	
CO3				2		3	3	3	2	3	1		
CO4				2		3	3	3	2	3			
CO5				2		3	3	3	2	3		1	

Course Assessment and Evaluation:

Sl.No	Components		Duration(min)	Marks Conducted for	Min Marks To pass the Course
1	Theory- Internal Assessment	IA-1	90	50	40
		IA-2	90	50	
	Total CIE			**100**	**40**
IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module					

SRI SIDDHARTHA ACADEMY OF HIGHER EDUCATION (SSAHE)

(Deemed to be University, Accredited A+ Grade by NAAC)

SRI SIDDHARTHA INSTITUTE OF TECHNOLOGY, TUMAKURU - 572105

(A Constituent College of SSAHE, Accredited by NBA, New Delhi in Tier - 1)

Scheme of Teaching and Examination for NEP 160 Credits (2025 Scheme)

4TH SEMESTER **ACADEMIC YEAR: 2026-27**

Sl. No.	Category and Course Code		Course Title	Teach. Dept.	Teaching Hrs. / Semester					Credits	Evaluation Marks			SEE Hrs.
					L	T	P	S	Total		CIE	SEE	Total	
1	ASC	25MA401	**Operations Research**	MA	42	-	-	48	90	3	50	50	100	3
2	PCC	25ME402	**Mechanical Measurements and Metrology**	ME	42	-	-	48	90	3	50	50	100	3
3	IPCC	25MEI403	**Fluid Mechanics And Fluid Machinery**	ME	42	-	28	50	120	4	50	50	100	3
4	PCC	25ME404	**CAD / CAM & CIM**	ME	42	-	-	48	90	3	50	50	100	3
5	PCC	25ME405	**Metal Casting & Forming Processes**	ME	42	-	-	48	90	3	50	50	100	3
6	PCC	25MEL406	**Mechanical Measurements and Metrology Laboratory**	ME	-	-	28	2	30	1	50	50	100	3
7	AECL	25MEAEL407	**CAM/CIM Laboratory**	ME	14	-	28	18	60	2	50	50	100	3
8	HSS	25CI408	**Constitution of India**	Any	14	-	-	16	30	1	100	-	100	-
9	EEC	25SK409	**Skill Development - 1**	T&P	14	-	-	16	30	1	100	-	100	-
Total					**252**	**-**	**84**	**294**	**630**	**21**	**550**	**350**	**900**	**-**
For lateral entry students only														
1	HSS	25EN410	**Environmental Studies**	Any	14	-	-	16	30	1	100	-	100	-

L: Lecture, T: Tutorial, P: Practical / Drawing, S: Student Academic Activity. C – Credits, CIE: Continuous Internal Evaluation, SEE: Semester End Examination

Department: Mathematics			Semester:	IV
Course: OPERATIONS RESEARCH				
Course Code:	25ME401	L – T – P - S:	42-0-0-48=90 (In Hrs)	
			Credits:	3

Sl. No	Course Objectives
This course will enable the students to :	
1	To equip the students with the knowledge based on OR models for problem solving and decision making situations in organizations.
2	To develop mathematical model for interactive decision making situations, where competitors are involved under conditions of conflict.
3	Construct the required activities in an efficient manner so as to complete it on or before a specified time limit and at minimum cost.
4	To make the student analyze the real time problems as operational research models and decide upon optimal solution.

Module	Description	Hrs
I	**Introduction of OR**, Definition of OR, Scope of OR, Application areas of OR, Phases of OR, Characteristics and limitations of OR, Linear Programing Problems (LPP) by graphical methods. **Simplex Methods**: Standard form of an LPP, Slack & Surplus Variables, Artificial variable or Big M Method.	**9**
II	**Transportation Problems:** Formulation of transportation problems, types, initial basic feasible solution using Penalty methods, initial basic feasible solution using North-West Corner rule, Vogel's Approximation method and optimal solution by MODI method, Degeneracy in transportation problems, Applications of Transportation problems, Concept for maximization cases. **Assignment problems**: Formation, Types, solution by Hungarian Method, Unbalanced problems, Comparison b/w Transportation &Assignment problems Application to maximization cases.	**9**
III	**Travelling salesman Problems**: Problems faced by travelling salesmen's and routing of machine operations by operators. **Sequencing:** Basic assumptions, sequencing 'N' jobs on single machine using priority rules, sequencing using Johnson's rule-'N' jobs on 2 machines, 'N' jobs on 3 machines.	**8**
IV	**Game Theory:** Definition, Pure Strategy problems, Formation of games, types, solution of games with saddle point, Graphical method of solving mixed strategy games, Dominance and Modified Dominance rule for solving mixed strategy games.	**8**
V	**PERT and CPM** Techniques: Introduction, Construction of networks, Fulkerson's rule for numbering the nodes, AON and AOA diagrams, definitions, different floats. Critical path method to find the expected completion time of a project. PERT networks, determining the probability of completing a project, predicting the completion time of project; Cost analysis in networks. Crashing of networks- Problems.	**8**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
I	Operations Research	Prem Kumar Gupta, Ds Hira	S Chand & Company Pvt Ltd (Publisher), 2016, Edition7, ISBN-139788121902816
II	Operations Research	Kalavathy S	Vikas Publishing House Pvt Ltd (Publisher), 2016, Edition4, ISBN-139789325963474
III	Operations Research An Introduction	Hamdy A. Taha	Pearson Ltd (Publisher), 10th Edition, Global Edition, ISBN 13: 978-1-292-16554-7

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Operations Research Theory & Applications	JK Sharma	Laxmi Publications (P) Ltd (Publisher), 2016, Edition6, ISBN-z39789385935145
2	Operations Research Theory Methods & Applications	SD Sharma	Kedar Nath Ram Nath (Publisher), 2017, Edition18, ISBN-139789380803388

E Resource's Links: (for self-study)

1. https://nptel.ac.in/courses/110106062
2. https://nptel.ac.in/courses/110106062
3. https://nptel.ac.in/courses/110106062
4. http://www.digimat.in/nptel/courses/video/111107128/L01.html

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	Understand the basic concepts LPP, Simplex method, Transportation, Assignment, Sequencing, Game theory, PERT and CPM.
CO2	Apply the fundamental knowledge of LPP, the Simplex method, Transportation and Assignment models, Sequencing, Game Theory, and project management tools like PERT and CPM to engineering problems.
CO3	Analyze the complex engineering problems using appropriate Mathematical techniques the Simplex method, Transportation and Assignment models, Sequencing, Game Theory, and project management tools like PERT and CPM.
CO4	The student will be able to examine optimal solutions using techniques such as LPP, the Simplex method, Transportation and Assignment models, Sequencing, Game Theory, and project management tools like PERT and CPM.
CO5	Enhance the comprehensive understanding of Transportation and Assignment problems, Sequencing, Game Theory, and project management tools like PERT and CPM gained to demonstrate the problems arising in real word situations.

Course Articulation Matrix

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	3	1								1	-	-
CO2	2	3	1	1							1	-	-
CO3	2	3	1	1	2						1	-	-
CO4	1	2	1	1	1						1	-	-
CO5	2	1		1	1						1	-	-

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks						
Sl. No	**Components**		**Duration (min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory- Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL1	-	25	5	4
		ABL2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module

Activity Based Learning(ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program(1 week/online or offline), ……Etc

Semester End Examination(SEE) : Max 100 Marks, Duration: 3 Hrs

The question paper shall contain **10 full questions (20 marks each),** with **2 questions from each of the 5 Modules,** ensuring **100% syllabus coverage.** Students shall answer **any 5 full questions** with **1:1 choice**. Each question may have a **maximum of 4 sub-divisions**.

Minimum passing Marks : 40% of Max Marks

Department: MECHANICAL ENGINEERING			Semester:	IV
Course: MECHANICAL MEASUREMENTS AND METROLOGY				
Course Code:	25ME402	L – T – P - S:	42-0-0-48=90 (In Hrs.)	
			Credits:	3

Sl. No	Course Objectives
This course will enable the students to :	
1	**Apply** the concepts to learn in metrology for applying the same in the selection of appropriate slip gauges, systems, application and standards.
2	**Apply** the concepts of limits, fits and tolerance during selective assembly and interchangeability in actual manufacturing process.
3	**Understand** and apply the concepts of measurement systems for checking accuracy, precision and calibration procedure of measuring instruments.
4	**Understand** the basic working principles of different transducers, intermediate and modifying devices and terminating devices, force, torque, pressure, temperature and strain.

Module	Description	Hrs
I	**Introduction to Metrology**: Definition, objectives and concept of metrology, need of inspection, principles, process, methods of measurement, errors in measurement. System of measurement, Material Standard, sub divisions of standard, digital metrology and smart measurement system, calibration of end bars (Numerical problems), Industry 4.0 Applications in Metrology. **Linear Measurement:** Slip gauges- Indian standards on slip gauges, wringing of slip gauges, uses slip gauges, care of slip gauges, problems on building of slip gauges (M87, M112). **Text 1:** Ch. **1, 2, 3 Text 2:** Ch. **1, 2, 3**	**9**
II	**System of Limits, Fits, Tolerance and Gauging:** Definition of tolerance, Principle of interchangeability and selective assembly, concept of limits of size and tolerances, compound tolerances, accumulation of tolerances, definition of fits, types of fits. Hole basis system and Shaft basis system. Classification of gauges, brief concept of design of gauges (Taylor's Principle), Types of gauges: plain plug gauge, ring gauge, numerical problems on limits, fits, tolerances and deviations. **Text 1:** Ch. **3, 4 Text 2:** Ch. **4, 5**	**9**
III	**Measurement systems**: Definition, requirements of measurement, significance of measurement, fundamental methods of measurement, generalized measurement system, definitions and basic concepts of accuracy, precision, calibration, threshold, sensitivity, hysteresis,	**8**

	repeatability, linearity, loading effect, Errors in measurement, classification of errors. **Transducers:** Introduction to Transducers, primary and secondary transducers, classification of transducers, advantages of each type transducers. **Text 1:** Ch. **2, 3, 14 Text 2:** Ch. **8, 9**	
IV	**Comparators:** Introduction to comparators, characteristics, classifications. Mechanical comparators: Johnson's Mikrokator, Sigma comparators, dial gauge, LVDT, optical comparator, Ziess-ultra optometer, pneumatic comparator-Solex pneumatic gauge. **Intermediate modifying and terminating devices**: Mechanical systems, Cathode ray oscilloscope, Oscillographs. **Text 1:** Ch. **15, 16 Text 2:** Ch. **6, 10**	**8**
V	**Measurement of Force, Torque and Pressure:** Force measurement, Mechanical dynamometer (Prony brake), Hydraulic dynamometer, Pressure measurement, Bridgeman gauge, Piranigauge,McLeod gauge. **Temperature and Strain measurement**: Resistance thermometers, thermocouple, law of thermocouple, materials used for construction, total radiation pyrometers, optical pyrometer. Strain measurements, strain gauge, preparation and mounting of strain gauges, gauge factor, methods of strain measurement. **Text 1:** Ch. **9, 10, 11 Text 2:** Ch. **11, 12, 13**	**8**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Mechanical Measurements	Beckwith Marangoni and Lienhard	Pearson Education, 6th Edition, 2006.
2	Engineering Metrology,	R.K. Jain	Khanna Publishers, Delhi, 2009.

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Mechanical Measurements	R.K. Jain	Khanna Publishers, 1994
2	Engineering Metrology	Gupta I.C.	Dhanpat Rai Publications,2016.
3	Engineering Metrology and Measurements,	N.V.Raghavendra & L.Krishnamurthy,	Oxford University Press,2018.

E Resource's Links: (for self-study)

1. https://nptel.ac.in/courses
2. https://swayam.gov.in
3. https://www.vlab.co.in
4. https://www.npl.co.uk

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	**Articulate** the fundamental concepts and objectives of metrology, including international standards, classification of measurement systems, error analysis, and the critical role of inspection in engineering practice.
CO2	**Demonstrate** proficiency in the selection and operation of precision linear and angular measurement devices (e.g., slip gauges, autocollimators, comparators), and **evaluate** their application in maintaining dimensional accuracy in engineering components.
CO3	**Differentiate** among various fits, tolerances, and gauge types, and **implement** appropriate measurement and gauging systems for mechanical assemblies, incorporating the principles of interchangeability and selective assembly.
CO4	**Apply analytical methods** to the measurement of screw threads and gears, and **diagnose** sources of measurement error by employing appropriate standards and metrological terminology.
CO5	**Integrate** mechanical and electronic transducers, intermediate devices, and sensors into experimental setups to **quantify** force, torque, pressure, temperature, and strain, and **interpret** measurement data for engineering problem-solving.

Course Articulation Matrix

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	1	2	1	1	1	1	1	2	1		
CO2	3	3	2	3	3	1	1	1	2	2	1		
CO3	3	2	2	2	2	1	1	1	2	2	1		
CO4	3	3	2	3	2	1	1	1	1	2	1		
CO5	3	3	3	3	3	1	1	1	2	2	2		

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks						
Sl. No	**Components**		**Duration(min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory- Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL1	-	25	5	4
		ABL2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module **Activity Based Learning(ABL):** Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz(Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program(1 week/online or offline), ……Etc
Semester End Examination(SEE) : Max 100 Marks, Duration: 3 Hrs The question paper shall contain**10 full questions (20 marks each),** with **2 questions from each of the 5 Modules,** ensuring**100% syllabus coverage.** Students shall answer **any 5 full questions** with **1:1 choice**. Each question may have a **maximum of 4 sub-divisions**. **Minimum passing Marks : 40% of Max Marks**

Department: Mechanical Engineering			Semester:	IV
Course: FLUID MECHANICS AND FLUID MACHINERY				
Course Code:	25MEI403	L – T – P - S:	42-0-28-50=120	(In Hrs)
			Credits:	4

Sl. No	Course Objectives
This course will enable the students to :	
1	**Understand** the fundamental properties of fluids and the principles governing fluid pressure, hydrostatic forces, buoyancy, and flotation.
2	**Understand** the behaviour of fluid flow using the concepts of fluid kinematics and fluid dynamics.
3	**Evaluate** fluid flow through pipes and flow measurement devices, including the estimation of discharge and energy losses.
4	**Explore** the construction, working principles, and performance characteristics of hydraulic turbines and centrifugal pumps.
5	**Apply** fluid mechanics principles for solving engineering problems related to hydraulic machinery, piping systems, and mechanical engineering applications.

Module	Description	Hrs
I	**Properties of Fluids**: Introduction, basic properties of fluids like Density, Specific weight, Specific volume, Specific gravity, Viscosity, Kinematic viscosity, Newton's law of viscosity, Variation of viscosity with temperature, Classification of fluids, Surface tension and capillarity, Vapour pressure and cavitation. Simple Numerical problems. **Text 1: Chapter 1: 1.1, 1.2, 1.3, 1.5, 1.6, 1.7** **Pressure and its Measurement:** Fluid Statics, Fluid pressure at a point, Pascal's law, pressure variation in a static fluid, Absolute, gauge, atmospheric and vacuum pressures, simple manometers, differential manometers and inverted manometers, Simple Numerical problems. **Text 1: Chapter 2: 1.2, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7.**	**8**
II	**Hydrostatics:** Total pressure and center of pressure on vertical plane surface submerged in liquid and inclined plane surface submerged in liquid, Simple Numerical problems. **Text 1: Chapter 3: 3.1, 3.2, 3.3, 3.4, 3.5.** **Buoyancy:** Buoyancy, Archimedes Principle, center of buoyancy, metacenter and metacentric height, conditions of equilibrium of floating bodies, determination of Metacentric height experimentally and theoretically, Numerical problems. **Text 1: Chapter 4: 4.1, 4.2, 4.3, 4.4, 4.5, 4.6, 4.7, 4.8.**	**9**
III	**Fluid Kinematics:** Introduction, types of flow, Continuity equation in three dimensions (Cartesian co-ordinate system only), Velocity and acceleration. Stream function and velocity potential function. Simple	**8**

	Numerical problems. **Text 1: Chapter 5: 5.1, 5.2, 5.3 5.4, 5.5, 5.6, 5.7, 5.8.** **Fluid Dynamics:** Introduction, Equations of motion, Euler's equation of motion Bernoulli's equation from Euler's equation, Bernoulli's equation for real fluids and simple numerical problems. **Text 1: Chapter 7: 7.1, 7.2, 7.3, 7.4, 7.5.**	
IV	**Fluid flow measurements**: Introduction, Concept of fluid flow measurement, Principle and derivation of expression for discharge through - Venturimeter, Orifice meter, Pitot tube, rectangular and triangular notches, Numerical problems. **Text 1: Chapter 8: 8.1, 8.2, 8.3, 8.4, 8.5.** **Flow through pipes**: Frictional loss in pipe flow, Darcy Equation for loss of head due to friction in pipes, Chezy's equation for loss of head due to friction in pipes, Hydraulic gradient and total energy line. Minor losses in the pipe, Simple Numerical problems. **Text 1: Chapter 9: 9.1, 9.2, 9.3, 9.4, 9.5, 9.6, 9.7.**	**8**
V	**Hydraulic Turbines:** Classification; Euler's Turbine Equation, Heads and efficiency of hydraulic turbine, Pelton Turbine-velocity triangles, Design parameters, turbine efficiency, volumetric efficiency; Francis turbine – velocity triangles, runner shapes for different blade speeds, Design parameters of Francis turbine; Function of a draft tube, types of draft tubes; Simple Numerical problems. **Text 2: Chapter 5.** **Centrifugal Pumps:** Working, Classification, Heads and efficiency of centrifugal pump, Types of vane shapes, Theoretical head – capacity relationship; Effect of blade discharge angle on performance; multistage centrifugal pumps, minimum starting speed, priming, cavitation, NPSH, Simple Numerical problems. **Text 2: Chapter 6.**	**9**

Lab Content

Sl. No	Experiment Description
1	Determination of Meta-centric Height by experimental Method.
2	Calibrate and determine the co-efficient of discharge through Venturimeter.
3	Calibrate and determine the co-efficient of discharge through orifice meter.
4	Calibrate and determine the co-efficient of discharge through V-notch.
5	Calibrate and determine the co-efficient of discharge through rectangular notch.
6	Determine the co-efficient of friction and loss of head in flow through pipes.
7	Determine minor losses of energy in flow through pipes.
8	Determination of force developed by impact of jets on vanes.
9	Performance testing of Turbines. 1. Pelton wheel. 2. Francis Turbine.
10	Performance testing of Centrifugal pumps.

Textbooks:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Fluid Mechanics and Hydraulic Machine	Dr. Bansal.R.K,	Lakshmi Publications, 2018
2	Turbomachines	Shankar Nag G. L. and Keerthi Kumar N.	1st Edition (2019), Cengage Learning India Pvt. Ltd.
3	Fluid Mechanics, Fundamental & Applications,	Yunus A Cenegel, John M Cimbala,	Tata McGraw Hill, 2006.

Reference Books:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Fluid Mechanics	John F. Douglas, Janul and M. Gasiosek and john A Swaffield,	Pearson Education Asia, 5th ed., 2006
2	Fluid Mechanics and Fluid Power Engineering,	D.S. Kumar.	Kataria and Sons, 2004.
	Hydraulics and Fluid Mechanics	Modi and Seth	Standard Book House, 22nd Edition, Delhi.

E Resource's Links: (for self-study)

1. www.youtube.com/watch?v=lJSUeEqGNY0&list=PLwdnzlV3ogoV-ATGY2ptuLS9mwLFOJoDw
2. www.youtube.com/watch?v=fa0zHI6nLUo&list=PLbMVogVj5nJTZJHsH6uLCO00I-ffGyBEm
3. www.youtube.com/watch?v=F_7OhKUYV5c&list=PL3F50D04B70A5B935
4. www.backbencher.club/fluid-mechanics-mechanical-engineering/

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	**Explain** the properties of fluids, pressure measurement techniques, hydrostatic forces, buoyancy and stability of floating bodies.
CO2	**Apply** the principles of fluid kinematics and fluid dynamics, including continuity, Euler's, and Bernoulli's equations to analyze fluid flow problems.
CO3	**Determine** discharge through flow measuring devices and **evaluate** head losses in pipe flow systems using appropriate equations and principles.
CO4	**Analyze** the performance characteristics of hydraulic turbines and centrifugal pumps and compute their operating parameters and efficiencies.
CO5	**Solve** engineering problems related to fluid properties, hydrostatics, fluid flow, hydraulic machines, and pipe systems using fundamental fluid mechanics

	concepts.

Course Articulation Matrix

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2											
CO2	3	3			1								
CO3	3	3	2	1	2								
CO4	3	3	2	1	2	1							
CO5	3	3	2	2	2	1					1		

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks						
Sl.No	**Components**		**Duration (min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory-Internal Assessment	IA-1	90	50	15	12
		IA-2	90	50	15	
2	Periodic Lab Assessment		-	30	10	4
3	Lab CIE		-	20	5	2
4	One Open Ended Experiment/ Project		-	10	5	2
	Total CIE			**160**	**50**	**20**

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module
Rubrics for Periodic Lab Assessment: Attendance (5M), Lab record (10 M), Conduction of Experiments(10M) and Viva(5M)-Total 30M
Lab CIE: Write-up(5M), Conduction/Execution/Demonstration(10M) and Viva(5M)-Total 20M

Semester End Examination(SEE) : **Max 100 Marks, Duration: 3 Hrs**
The question paper shall contain **10 full questions (20 marks each),** with **2 questions from each of the 5 Modules**, ensuring **100% syllabus coverage**. Students shall answer **any 5 full questions** with **1:1 choice.** Each question may have a **maximum of 4 sub-divisions.**
Minimum passing Marks : 40% of Max Marks

Department: Mechanical Engineering			Semester:	IV
Course: CAD/CAM & CIM				
Course Code:	25ME404	L – T – P - S:	42-0-0-48=90	(In Hrs)
			Credits:	03

Sl. No	Course Objectives
This course will enable the students to :	
1	To **introduce** the principles, concepts, and applications of CAD/CAM in modern manufacturing systems.
2	To **develop** the ability to design and generate CNC part programs for machining operations using standard programming techniques.
3	To **enable** students to apply the principles of automated assembly systems in manufacturing environments.
4	To **analyze** the fundamentals, configurations, and industrial applications of robots in automated manufacturing systems.

Module	Description	Hrs
I	**Introduction to CAD/CAM & CIM**: CAD/CAM & CIM Definition, operations in manufacturing, product life cycle and CAD/CAM in Automation, production concepts, Design process, Application of computers in design, Creating manufacturing database in CAD, CIM -lean production, facilities layout, types of layout, layout facilities for automated manufacturing, information process in manufacturing, Advantages and disadvantages of CAD , CAM& CIM. **Ref: Text 1 (Sec 1.1, 1.2),Text 2 (Sec 3.1, 3.2),Text 1 (Sec 10.1, 10.2, 14.1)**	**9**
II	**Automation:** definition, types of automation, levels of automation, automation strategies, mathematical model for production parameters, problems. **High volume production:** introduction, automated flow lines, flow line configurations, work part transport methods, work part transfer mechanism, buffer storage, need for buffer storage **Ref: Text 1 (Sec 3.1–3.5),Text 1 (Sec 16.1–16.6)**	**9**
III	**Automated Assembly Systems:** Automated assembly systems: Product design for automated assembly, types of automated assembly systems, Part feeding and delivery system-hopper, part feeder, selector, feedback, escapement and placement. **Automated Guide Vehicles:** introduction, automated material handling, types of material handling equipment, principles of material handling, selection of material handling equipment, Automated guide vehicle systems, vehicle guiding and routing. **Ref: Text 1 (Sec 17.1–17.4),Text 1 (Sec 9.1–9.3)**	**8**

IV	**CNC Machining entres:** Introduction to NC, NC Technology, importance of NC, NC coordinate system, NC motion control system, Basics of NC machine operation, Application, advantages and limitations of NC Systems. Introduction to CNC, need for CNC, CNCsystem, Need for CNC system, CNC machining centres(HMC &VMC),Advantages of CNC systems. **CNC Programming;** part programming, fundamental steps involved in development of part programming, Manual part programming for milling, drilling and turning. **Ref: Text 2 (Sec 8.1–8.5), Text 2 (Sec 9.1–9.6)** .	**8**
V	**ROBOTICS:** Introduction, Robotic Fundamentals, Physical configuration of robot, Robot motions, Joints in Robot, Technical features of robots, End effector, Robot sensors and Robot applications **Ref: Text 1 (Sec 7.1–7.6)**	**8**

Textbooks:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Automation, Production system & Computer Integrated manufacturing	M. P. Groover	Pearson India, 2nd edition, 2007.
2	CAD/CAM	Zeid	Tata McGraw Hill,2009.

Reference Books:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Computer Integrated Manufacturing,	J. A. Rehg and Henry. W. Kraebber.	Pearson India , 2nd Edition , 2000
2	Principles of Computer Integrated Manufacturing,	S. Kant Vajpayee	Prentice Hall India,2011

E Resource's Links: (for self-study)

1. https://media.dev.nptel.ac.in/content/mp4/112/104/112104289/MP4/mod01lec01.mp4
2. https://media.dev.nptel.ac.in/content/mp4/112/104/112104289/MP4/mod04lec17.mp4
3. https://media.dev.nptel.ac.in/content/mp4/112/104/112104289/MP4/mod05lec20.mp4
4. https://media.dev.nptel.ac.in/content/mp4/112/104/112104289/MP4/mod08lec32.mp4
5. https://media.dev.nptel.ac.in/content/mp4/112/104/112104289/MP4/mod08lec31.mp4

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	**Explain** the principles and applications of CAD, CAM, and CIM, Automation and Robotics in modern manufacturing systems
CO2	**Analyze** different types of automation and develop mathematical models for production systems.

CO3	**Evaluate** the working and design principles of automated assembly systems,automated guided vehicle (AGV) systems and CNC machining.
CO4	**Demonstrate** knowledge of CNC machining centers and develop manual part programs for basic machining operations
CO5	**Describe** the fundamentals of robotics, including configurations, joints, sensors, and end effectors, and identify suitable applications of robots in manufacturing.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2					3		1				2	
CO2	1	3			3					2	2	3	
CO3		1	2		2	2				2	2	2	
CO4			3					3	2		3	3	
CO5	3		3								2	3	

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks						
Sl. No	**Components**		**Duration (min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory- Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL1	-	25	5	4
		ABL2	-	25	5	
Total CIE				150	50	20

IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module

Activity Based Learning(ABL): Online course with certification, Hobby project, Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program(1 week/online or offline), ……Etc

Semester End Examination(SEE) : Max 100 Marks, Duration: 3 Hrs

The question paper shall contain **10 full questions (20 marks each),** with **2 questions from each of the 5 Modules,** ensuring **100% syllabus coverage.** Students shall answer **any 5 full questions** with **1:1 choice**. Each question may have a **maximum of 4 sub-divisions**. **Minimum passing Marks : 40% of Max Marks**

Department: Mechanical Engineering			Semester:	IV
Course: METAL CASTING AND FORMING PROCESSES				
Course Code:	25ME405	L – T – P - S:	42-0-0-48=90	(In Hrs)
			Credits:	3

Sl. No	Course Objectives
This course will enable the students to :	
1	**Understand** the fundamentals of manufacturing processes, casting operations, pattern making, core making, gating systems, risers, and casting defects.
2	**Explain** various moulding processes, metal mould casting methods, melting furnaces, and their applications in manufacturing industries.
3	**Analyze** metal forming processes such as forging, rolling, sheet metal operations, drawing, and extrusion, including their advantages, limitations, and defects.
4	**Evaluate** and **Select** appropriate joining processes, including soldering, brazing, and welding techniques, for engineering manufacturing applications.

Module	Description	Hrs
I	**Introduction to manufacturing processes:** Definition, Classification of manufacturing processes, introduction to casting, steps involved in making a casting, procedure for making casting, advantages, disadvantages and application of casting process. **Patterns:** Definition, function of pattern, materials used for pattern, pattern allowances and classification of patterns. **Cores:** Definition, types of cores **Concept of gating and risers:** principle of gating system, types of gates, types of risers, casting defects. **Text 1: 1.1, 2.1. 3.1, 4.1, Text 3: 1.1**	9
II	**Study of important moulding process:** sweep mould, shell mould, investment mould. **Casting using metal moulds:** Gravity die casting, pressure die casting, squeeze casting and continuous casting processes. **Melting furnaces:** Classification of furnaces, Resistance furnace, Coreless induction furnace, constructional features & working principle of cupola furnace. **Text 1: 5.1, 6.1**	9
III	**Mechanical Working of Metals:** Introduction to metal forming processes & classification of metal forming processes, advantages and disadvantages of metal working process. **Forging:** Classification of forging processes, defects in forging,	8

	advantages & disadvantages of forging. **Sheet Metal Operations:** Blanking, punching (piercing), Deep drawing, Trimming, and Shearing **Text 2: 4.1**	
IV	**Rolling:** classification of Rolling process, Types of rolling mills, Rolling defects. **Drawing & Extrusion:** Drawing process, drawing equipment's, Extrusion of rod, types of extrusion. **Joining process:** principle of soldering and brazing, Advantages and disadvantages of soldering, advantages and limitations of brazing, comparison of soldering and brazing. **Text 1: 30.1, Text 2: 4.1**	8
V	**Welding:** Principle of welding, classification of welding process, Application of welding, Advantages and limitations of welding. Arc welding: Principle, Metal arc welding (MAW) or Flux Shielded Metal Arc Welding (FSMAW), Tungsten Inert Gas Welding (TIG), Metal insert gas (MIG) welding, Gas welding, oxy-Acetylene welding, and types of flames produced in gas welding process. **Special type of welding:** Spot welding, Seam welding, Friction welding, Explosive welding, Laser welding and Electron beam welding. **Text 1: 21.1, 22.1, 23.1, 24.1, 25.1, 26.1**	8

Textbooks:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Manufacturing Process - I	Dr. K. Radhakrishna	Sapna Book House, 5th Revised Edition 2009.
2	Manufacturing Process - II	Dr. K. Radhakrishna	Sapna Book House, 5th Revised Edition 2009.
3	Principles of metal casting	Rechard W. Heine, Carl R. Loper Jr., Philip C. Rosenthal, Tata	McGraw Hill Education Private Limited Ed.1976.

Reference Books:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Process and Materials of Manufacturing	Roy A Lindberg	Pearson Education 4th Ed. 2006.
2	Manufacturing Technology	SeropeKalpakjian, Steuen. R. Sechmid,	Pearson Education Asia, 5th Ed. 2006.

E Resource's Links: (for self-study)

1. https://nptel.ac.in/courses/112107215
2. https://onlinecourses.nptel.ac.in/e-learning/preview/noc21_me87
3. https://nptel.ac.in/courses/112103263
4. https://nptel.ac.in/courses/112107219

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	**Explain** the fundamentals of manufacturing processes, casting operations, patterns, cores, gating systems, risers, and casting defects.
CO2	**Describe** various moulding methods, metal mould casting processes, and melting furnaces used in foundry practice.
CO3	**Illustrate** the principles, classifications, advantages, limitations, and defects of metal forming processes such as forging and sheet metal operations.
CO4	**Compare** rolling, drawing, extrusion, soldering, and brazing processes and **Select** suitable manufacturing processes for engineering applications.
CO5	**Analyze** different welding processes, including arc welding, gas welding, and special welding techniques, to **Recommend** suitable engineering applications.

Course Articulation Matrix

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	1		1								
CO2	3	2	2		1								
CO3	3	3	2		1								
CO4	3	3	2		1								
CO5	3	3	3	1	2								

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks						
Sl. No	**Components**		**Duration (min)**	**Marks Conducted for**	**Marks reduced to**	**40% Min Marks in each component to write SEE**
1	Theory- Internal Assessment	IA-1	90	50	20	16
		IA-2	90	50	20	
2	Activity Based Learning	ABL1	-	25	5	4
		ABL2	-	25	5	
Total CIE				150	50	20
IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module **Activity Based Learning(ABL):** Online course with certification, Hobby project,						

Simulations, Industrial Visits, Community-Based Learning, Case-study, Problem-Based Learning, Peer Teaching, web page design, student publications, Hackathons, Quiz (Leetcode, GitHub, Hackaranks and company specific quiz), Student Development program(1 week/online or offline), ……Etc
Semester End Examination(SEE) : Max 100 Marks, Duration: 3 Hrs The question paper shall contain **10 full questions (20 marks each),** with **2 questions from each of the 5 Modules,** ensuring **100% syllabus coverage.** Students shall answer **any 5 full questions** with **1:1 choice**. Each question may have a **maximum of 4 sub-divisions**. **Minimum passing Marks : 40% of Max Marks**

Department: Mechanical Engineering			**Semester:**	**IV**
COURSE: MECHANICAL MEASUREMENTS AND METROLOGY LABORATORY (Professional Core Laboratory - 2)				
Course Code:	**25 ME L406**	**L – T – P - S:**	**0-0-28-2=30**	**(In Hrs)**
			Credits:	**01**

Sl. No	**Course Objectives**
This course will enable the students to :	
1	**Understand** the principles of calibration and measurement of mechanical and thermal parameters using standard instruments.
2	**Develop** practical skills in the operation of metrology instruments for dimensional and angular measurements.
3	**Analyze** the accuracy of the obtained results and perform measurement, testing, and calibration experiments.
4	**Apply** metrology and measurement techniques in quality control and manufacturing applications.

Sl.no	**Experiments**
1	Calibration of Pressure Gauge
2	Calibration of Thermocouple
3	Calibration of LVDT
4	Calibration of Micrometer using slip gauges.
5	Calibration of Load cell
6	Measurement using Optical Projector /Toolmaker Microscope.
7	Measurement of angle using Sine Center / Sine bar/Bevel protractor
8	Measurement of gear tooth profile using gear tooth Vernier/Gear Tooth micrometer
9	Measurement of flatness by using monochromatic check light.

Text Books:

Sl No	**Text Book title**	**Author**	**Volume and Year of Edition**
1	Mechanical Measurements	Thomas G. Beckwith, Roy D. Marangoni and John H. Lienhard V	Pearson Education, **6th Edition, 200**
2	Engineering Metrology	R. K. Jain	Khanna Publishers, **20th Edition, 2022**

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Engineering Metrology and Measurements	N. V. Raghavendra and L. Krishnamurthy	Oxford University Press, 2nd Edition, 2021

E Resource's Links: (for self-study)

1. https://ocw.mit.edu/courses/mechanical-engineering
2. https://www.engineeringtoolbox.com
3. https://www.omega.com
4. https://www.keysight.com

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	**Calibrate** pressure gauges, thermocouples, LVDTs, load cells, and evaluate their performance characteristics.
CO2	**Perform** dimensional measurements using slip gauges, optical projectors, and toolmaker's microscopes with appropriate accuracy.
CO3	**Understand** angles, gear tooth parameters, and geometric features using standard metrology instruments and interpret the results.
CO4	**Conduct** experiments related to flatness measurement and analyze measurement data for engineering applications.
CO5	**Apply** suitable measurement and calibration techniques for quality inspection and industrial metrology applications.

Course Articulation Matrix

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	-	2	3	-	-	-	-	-	-		
CO2	3	3	-	2	3	-	-	-	-	-	-		
CO3	3	3	-	3	2	-	-	-	1	-	-		
CO4	2	3	-	3	2	-	-	-	1	-	-		
CO5	3	3	1	3	3	-	-	-	1	1	-		

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks			
Sl.No	**Components**	**Marks Conducted for**	**40% Min Marks in each components to write SEE**

1	Periodic Lab Assessment	30	12
2	Lab CIE	20	8
	Total CIE	50	20

Rubrics for Periodic Lab Assessment: Attendance (5M), Lab record (10M), Conduction of Experiments(10M) and Viva((5M)-Total 30M
Lab CIE: Write-up(5M), Conduction/Execution/Demonstration(10M) and Viva(5M)-Total 20M

Semester End Examination(SEE) : Max 50 Marks, Duration: 3 Hrs
Rubrics for Lab Examination: Write-up(8M), Conduction/Execution/Demonstration(35M) and Viva(7M)-Total 50M.
Minimum passing Marks:40% of Max Marks

Department: Mechanical Engineering			Semester:	IV
Course: CAM/CIM LABORATORY (Ability Enhancement Laboratory 2)				
Course Code:	25MEAEL407	L – T – P - S:	14-0-28-18=60 (In Hrs)	
			Credits:	02

Sl. No	Course Objectives
This course will enable the students to :	
1	To **familiarize** students with CAM software for the simulation of CNC turning, drilling, and milling operations.
2	To **provide** hands-on experience in CNC programming and the application of canned cycles for machining operations.
3	To **introduce** the design and modeling of robotic manipulators with different degrees of freedom using ROBO Analyzer software.
4	To **impart** knowledge of plant layout planning techniques and manufacturing system design for efficient material flow and resource utilization.

	Description	Hrs
PART A	1. Simulation of Turning operations using CAM simulation software 2. Simulation of Drilling operations using CAM simulation software 3. Simulation of Milling operations using CAM simulation software 4. Simulation of Canned Cycles for Milling operations using CAM simulation software 5. Simulation of Canned Cycles for Turning operations using CAM simulation software	**7+14**
PART B	6. Designing R-R-R(Revolute- Revolute - Revolute) type of ROBOT manipulators using ROBO Analyzer software 7. Designing P-R-P(Prismatic- Revolute - Prismatic) type of ROBOT manipulators using ROBO Analyzer software 8. Designing P-P-P(Prismatic- Prismatic- Prismatic) type of ROBOT manipulators using ROBO Analyzer software 9. Study of Plant layout using string diagram 10. Development of product layout for the given product using systematic layout planning principles. 11. Development of process layout for the given product using systematic layout planning principles. 12. Development of hybrid layout for the given product using systematic layout planning principles.	**7+18**

Textbooks:

Sl No	Textbook title	Author	Volume and Year of Edition
1	CAD/CAM	Zeid	Tata McGraw Hill,2009.
2	Introduction to Robotics	John Chraig	Perasons,4th Edition

3	Facilities Planning	James A. Tompkins, John A. White, Yavuz A. Bozer, and J.M.A. Tanchoco	4th Edition, Pearson Education.

Reference Books:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Computer Integrated Manufacturing,	J. A. Rehg and Henry. W. Kraebber.	Pearson India , 2nd Edition , 2000
2	Introduction to Robotics: Analysis, Control, Applications,	Saeed B. Niku	3rd Edition, Wiley India, 2021
3	Plant Layout and Material Handling	James M. Apple	Krieger Publishing Company

E Resource's Links: (for self-study)

1. https://media.dev.nptel.ac.in/content/mp4/112/104/112104289/MP4/mod01lec01.mp4
2. https://media.dev.nptel.ac.in/content/mp4/112/104/112104289/MP4/mod08lec32.mp4
3. https://www.roboanalyzer.com/tutorials.html
4. https://youtu.be/4vq0FKWYud8

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	**Develop** and **simulate** CNC machining operations such as turning, drilling, and milling using CAM simulation software.
CO2	**Generate** and verify CNC programs incorporating various canned cycles for turning and milling operations through simulation.
CO3	**Design** and model robotic manipulators with different degrees of freedom (2-DOF and 3-DOF) using robotic simulation software.
CO4	**Develop** plant layouts using string diagrams and cellular manufacturing concepts for efficient material flow.
CO5	**Design** product, process, and hybrid layouts using systematic layout planning principles to improve manufacturing efficiency.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	1	1	3							3	
CO2	3	3	2	2	3							3	
CO3	2	2	3	2	3				1			3	
CO4	2	3	2	3	3		1		1			2	
CO5	2	3	3	2	3		1		2	1	1	3	

Course Assessment and Evaluation:

This course consists of experimental and project-based learning.

- **Part A:** 5 weeks of experimental learning with 5 experiments. Students shall conduct the experiments individually. CIE for Part A shall be conducted in the 7th week.
- **Part B:** 5 weeks of project-based learning. A team of maximum 4 students shall carry out a project. The project shall be completed by the 13th week; CIE for Part B may be conducted in the 14th week.

Continuous Internal Evaluation (CIE): 50 Marks				
Sl. No	**Components**	**Marks Conducted for**	**Marks Reduced to**	**40% Min Marks in each component to write SEE**
CIE for PART A(Experimental Learning)				
1	Periodic Lab Assessment	30	15	6
2	Lab CIE	20	10	4
	Total CIE for Part A	50	25	10
Rubrics for Periodic Lab Assessment: Attendance (5M), Lab record (10 M), Conduction of Experiments(10) and Viva(5M)-Total 30M **Lab CIE: Write-up(5M), Conduction/Execution/Demonstration(10M) and Viva(5M)-Total 20M**				
CIE for PART B (Project based Learning)				
1	Lab CIE Part B	50	25	10
	Total CIE(Part A and Part B)	**100**	**50**	**20**
Rubric for Lab CIE Part B: Write-up (10M), Presentation(10M),Demonstration(20M) Viva(10M)- **Total 50 Marks**				
Semester End Examination (SEE) : 50 Marks, Duration: 3 Hrs				
PART A (25 Marks): **Rubrics for Part A: Write-up (5M), Conduction/Execution/Demonstration (15M) and Viva (5M) - Total 25M.** **PART B (25 Marks):** **Rubrics for Part B: Write-up (5M), Presentation/Demonstration (15M) and Viva (5M)-Total 25M.** **Minimum passing Marks: 40% in Part A and Part B each**				

Department: Mechanical Engineering			Semester:	IV
Course: CONSTITUTION OF INDIA				
Course Code:	25CI408	L – T – P - S:	14-0-0-16 =30 (In Hrs)	
			Credits:	1

Sl. No	Course Objectives
This course will enable the students to :	
1	**Explain** the salient features and Preamble of the Constitution of India, Fundamental Rights, and Fundamental Duties.
2	**Interpret** the relevance of Directive Principles of State Policy and the responsibilities of individuals towards society.
3	**Explain** the powers and functions of legislative bodies.
4	**Describe** Fundamental Duties, the electoral process, emergency provisions, and constitutional amendment procedures.

Module	Description	Hrs
I	Introduction, Meaning and definitions. Salient features, Sources, Constituent Assembly, Drafting Committee. Preamble to the constitution of India.	**2**
II	Fundamental rights under part III, scope, Limitations and important cases, Judicial and Excecutive	**2**
III	Relevance of directive principles of state policy, fundamental duties and its significance, different commissions of india, Finance Commission, five year plans, 73rd & 74th Constitutional Amendment	**3**
IV	Union Executives- President, Vice President, Prime minister, Speaker, Parliament, Supreme Court of India. State Executives – Governor, Chief Minister, (Upper and lower house) State legislature and High Court.	**3**
V	Constitutional Special Provisions for Scheduled Castes and Tribes, Women, Children and backward classes. Electoral process. Ethics : Introduction Scope of Ethics , History of Ethics, Methods of Ethics Different Approaches to the Study of Ethics, Division of Ethics	**4**

Textbooks:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Introduction to the Constitution of India (student edition)	Durga Das Basu	EEE, 19th/20th Edn.,2001
2	An Introduction to Constitution of India	MV Pylee	Volume-1 ,Vikas Publishing, 2002

Reference Books:

Sl No	Textbook title	Author	Volume and Year of Edition
1	An Introduction to Constitution of India	Brij kishore Sharma	Prentice-Hall of India, Volume-1, 2002
2	Constitution of India and Professional Ethics	V. Rajaram	Second Edition, New Age International Publication. 2011

E Resource's Links: (for self-study)

1. https://egyankosh.ac.in/bitstream/123456789/81071/1/Block 1.pdf

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	**Demonstrate** general knowledge and legal literacy required for competitive examinations.
CO2	**Explain** Fundamental Rights, Directive Principles of State Policy, and Fundamental Duties.
CO3	**Analyze** the roles and functions of the three main organs of the Constitution: Legislature, Executive, and Judiciary.
CO4	**Recall** the various Constitutional Amendments and provisions.
CO5	**Identify** the functions and structure of Union and State Executives.

Course Articulation Matrix

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1						2	1	3	3	2			
CO2		2			2				3	2	1		
CO3	1		1					3	2	2		1	
CO4				1				3	3	2			
CO5							1						

Course Assessment and Evaluation:

Sl.No	Components		Duration(min)	Marks Conducted for	Min Marks To pass the Course
1	Theory- Internal Assessment	IA-1	90	50	40
		IA-2	90	50	
	Total CIE			**100**	**40**
IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module					

Department: Mechanical Engineering			Semester:	IV
Course: Fundamentals of Aptitude and Employability Skills (SKILL DEVELOPMENT – 1)				
Course Code:	25SK409	L – T – P - S:	14-0-0-16 =30 (In Hrs)	
			Credits:	1

Sl. No	Course Objectives
This course will enable the students to :	
1	**Develop** fundamental quantitative aptitude skills required for problem-solving in placement assessments and competitive examinations.
2	**Apply** logical reasoning techniques to analyze patterns, relationships and solve reasoning problems systematically.
3	**Enhance** verbal communication and language proficiency through vocabulary development, comprehension and error analysis.
4	**Demonstrate** effective communication, professional ethics and employability skills required for career development and placement readiness.

Module	Description	Hrs
I	**Quantitative Aptitude Basics:** Number Systems, HCF and LCM, Percentages, Profit and Loss, Ratio and Proportion, Averages, Simple and Compound Interest, Time and Work, Time, Speed and Distance.	3
II	**Logical Reasoning Fundamentals:** Series Completion, Coding-Decoding, Blood Relations, Direction Sense, Ranking and Ordering, Puzzles, Seating Arrangements.	3
III	**Verbal Ability Basics:** Vocabulary Building, Synonyms and Antonyms, One-word Substitutions, Idioms and Phrases, Sentence Correction, Error Detection, Reading Comprehension Basics.	3
IV	**Communication Skills:** Fundamentals of Communication, Listening Skills, Public Speaking, Presentation Skills, Group Discussion Introduction, Confidence Building Activities.	3
V	**Employability Awareness:** Placement Process Overview, Career Planning, Industry Expectations, Professional Ethics.	2

Textbooks:

Sl No	Textbook title	Author	Volume and Year of Edition
1	Quantitative Aptitude for Competitive Examinations	R. S. Aggarwal	Revised Edition, S. Chand, 2024

2	A Modern Approach to Verbal & Non-Verbal Reasoning	R. S. Aggarwal	Revised Edition, S. Chand, 2024
3	Soft Skills: Know Yourself and Know the World	Dr. Alex K.	3rd Edition, S. Chand, 2022

Reference Books:

Sl No	Textbook title	Author	Volume and Year of Edition
1	How to Prepare for Quantitative Aptitude for CAT	Arun Sharma	10th Edition, McGraw Hill, 2024
2	Word Power Made Easy	Norman Lewis	Revised Edition, 2023
3	Business Communication Today	Bovee, Thill & Chatterjee	15th Edition, Pearson, 2023

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	**Illustrate** employability awareness, professional ethics and career planning strategies for placement preparedness.
CO2	**Solve** quantitative aptitude problems involving arithmetic, percentages, ratios, interests and work-related concepts.
CO3	**Demonstrate** proficiency in verbal ability through vocabulary usage, grammar correction and reading comprehension.
CO4	**Exhibit** effective communication skills through presentations, discussions and interpersonal interactions.
CO5	**Apply** logical reasoning techniques to analyze and solve analytical and pattern-based problems.

Course Articulation Matrix

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	-	-	-	-	-	-	-	-	-	-	-
CO2	2	3	1	-	-	-	-	-	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	2	3	-	-	-
CO4	-	1	-	-	-	-	-	2	3	3	-	-	-
CO5	-	-	-	-	-	1	-	3	2	2	1	-	-

Course Assessment and Evaluation:

Sl.No	Components		Duration(min)	Marks Conducted for	Min Marks To pass the Course
1	Theory- Internal Assessment	IA-1	90	50	40
		IA-2	90	50	
	Total CIE			**100**	**40**
IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module					

ENVIRONMENTAL STUDIES

For lateral entry students only

Department: Mechanical Engineering			Semester:	IV
Course: ENVIRONMENTAL STUDIES				
Course Code:	25EN410	L – T – P - S:	14-0-0-16 =30 (In Hrs)	
			Credits:	1

Sl. No	Course Objectives
1	Explain the fundamental concepts of environmental science, sustainability, ecosystems, natural resources, and biodiversity to develop environmental awareness.
2	Analyze the sources, impacts, and control measures of environmental pollution and waste management practices.
3	Apply environmental laws, ethics, and sustainable engineering practices to address environmental challenges responsibly.
4	Develop disaster management plans by integrating preparedness, mitigation, response, and recovery measures while evaluating global environmental issues.

Module	Description	Hrs
I	**Introduction**: Environment - Components of Environment, Atmosphere (Structure & Composition), Hydrological cycle, Ecosystem: Types & Function of Ecosystem, Effects of human activities on environment by Agriculture, Industry and urbanization, Environmental Impact Assessment (**EIA**), **Refer text book 1: Ch-1,2: text book 2: Ch-1,2** **Reference book: Ch-1,2 from book 1 & 2**	**3**
II	**SDG:** Sustainable Development goals **(SDG's),** Introduction To Sustainable Development Definitions and principles of Sustainable Development – History, Millennium Development Goals: Status (global and Indian) – Sustainable Development Goals (SDGs) and their relevance to Engineering, **Natural Resources**- types of natural resources, Water resources – Surface and subsurface water, Quality aspects, Drinking water quality standards. **Refer text book 1: Ch-3: text book 2: Ch-3** **Reference book: Ch-3,4 from book 1 & 2 and Ch-1 from book 3**	**3**
III	**Environmental Pollution** (Causes, effect and control) – Air pollution, Surface Water Pollution, Ground water pollution, Noise pollution, Soil Pollution, Environmental Acts & Regulations to control Pollution, Integrated Water Resources Management (IWRM) & IoT-Based Monitoring, smart waste management, AI, Machine Learning, Robotics, GIS and Remote Sensing application in Environmental studies. **Refer text book 1: Ch-5: text book 2: Ch-4** **Reference book: Ch-5,6,7 from book 1 & 2**	**3**
IV	**Global Environmental Issues:** Global Warming, Acid rain & Ozone layer depletion (Causes, effect and control), Climate change, Carbon trading, Net zero emission **Energy:** Types, alternative sources of Energy, **Smart**	**3**

	cities and sustainable urban planning, Project Management in Smart Cities-Case studies on project management of smart cities **Refer text book 1: Ch-6: text book 2: Ch-5** **Reference book: Ch-8,9 from book 1 & 2 and Ch-2,3,4 from book 3**	
V	**Waste Management & Disposal:** E-Waste, Biomedical Waste, Hazardous waste (Management & Disposal), Plastic waste, National Environmental policy, **Disaster management Plan:** Types of disasters and its impacts, awareness and preparedness, Disaster Risk Management In India, NDMA, NDRF, SDRF. Develop plans to mitigate disasters **Refer text book 1: Ch-6,7: text book 2: Ch-6** **Reference book: Ch-10, 11 from book 1 & 2 and Ch-8 from book 3**	**2**

Text Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Environmental Studies for Undergraduate Courses	Erach Bharucha	Universities Press, 3rd Edition, 2023.
2	Environmental Science and Engineering	P. Anandan and R. Kumaravelan	Scitech Publications, 2026 Edition.

Reference Books:

Sl No	Text Book title	Author	Volume and Year of Edition
1	Environmental Studies	R Rajagopalan	From Crisis to Cure", 4th Edition, 2022, Oxford University Press
2	Environmental Studies	Benny Joseph	3rd Edition, 2023, McGraw Hill Education
3	Sustainable Construction and Green Infrastructure	Charles J. Kibert	5th Edition, 2024

Course Outcomes:

Course outcome: After the successful completion of the course, the student will be able to	
CO1	Explain the fundamental concepts of environmental science, ecosystems, natural resources, biodiversity, and their significance in sustainable development.
CO2	Analyze the causes, impacts, and control measures of environmental pollution, climate change, and waste management practices..
CO3	Evaluate environmental issues related to water resources, energy resources, and environmental conservation using scientific and engineering principles.
CO4	Apply environmental laws, sustainability principles, and ethical responsibilities to develop environmentally responsible engineering solutions and promote sustainable lifestyles.
CO5	Develop disaster management plans incorporating disaster preparedness, mitigation, response, and recovery measures to minimize the impacts of natural and human-induced disasters.

Course Articulation Matrix

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1					1							
CO2						2	2				1		
CO3		2				1							
CO4			1			2					1		
CO5						1							

Course Assessment and Evaluation:

Sl.No	Components		Duration(min)	Marks Conducted for	Min Marks To pass the Course
1	Theory- Internal Assessment	IA-1	90	50	40
		IA-2	90	50	
	Total CIE			**100**	**40**
IA-1 Syllabus: 2 ½ Module , IA-2 Syllabus: 2 ½ Module					

Signature of the HoD

Signature of the Dean (Academic Affairs)